Documentatie

Protocoale folosite:

- Hostname
- Vlan
- Port-channel
- Etherchannel
- RSTP
- LACP
- Access/Trunk
- Portfast
- BPDU guard
- Encapsulation

Parola Firewalls cisco

De adaugat:

```
interface range Ethernet0/2 - 3
 switchport port-security
 switchport port-security maximum 2
 switchport port-security violation shutdown
 switchport port-security mac-address sticky
```

NTP (ierarhic — important pentru loguri medicale) + **Syslog** centralizat
De verificat redundanta la edge-routers. Daca pica unul sa se ruteze pe celalalt.

```
vlan 10
 name Zabbix
vlan 30
 name AAA
vlan 99
 name Native
```

```
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
```

```
spanning-tree mode rapid-pvst
spanning-tree vlan 5,10,20,30,40 priority 4096
```

```
spanning-tree portfast
spanning-tree bpduguard enable
```

```
interface range Ethernet1/0 - 1
 channel-group 1 mode active
interface Port-channel1
 switchport mode trunk
```

```
switchport port-security
switchport port-security maximum 2
switchport port-security violation shutdown
switchport port-security mac-address sticky
```

```
ip dhcp snooping
ip dhcp snooping vlan 10,20
interface Port-channel1
 ip dhcp snooping trust
```

```
standby version 2
standby 5 ip 192.168.5.1
standby 5 priority 150
standby 5 preempt
```

```
router ospf 1
 router-id 1.1.1.1
 network 10.0.2.0 0.0.0.3 area 0
 default-information originate
```

```
ip nat inside
ip nat outside
access-list 1 permit 10.0.0.0 0.255.255.255
ip nat inside source list 1 interface Ethernet0/2 overload
```

```
crypto isakmp policy 10
 encryption aes 256
 hash sha256
 authentication pre-share
 group 14
```

```
crypto ipsec transform-set TS-GRE esp-aes esp-sha256-hmac
interface Tunnel11
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/2
 tunnel destination 203.0.113.10
 tunnel protection ipsec profile IPSEC-PROF
```

```
nameif inside
 security-level 100
nameif outside
 security-level 0
```

```
access-list OUTSIDE_IN extended permit ospf host 10.0.2.1 any
access-list OUTSIDE_IN extended permit icmp 192.168.0.0 255.255.0.0 any
access-group OUTSIDE_IN in interface outside
```

```
class-map TCP_CLASS
 match access-list TCP_BYPASS
policy-map global_policy
 class TCP_CLASS
  set connection advanced-options tcp-state-bypass
```

```
ip domain name spital.local
crypto key generate rsa modulus 2048
ip ssh version 2
```

```
username gns3 privilege 15 secret gns3
line vty 0 4
 login local
 transport input ssh
```

```
subnet 10.1.10.0 netmask 255.255.255.0 {
		range 10.1.10.50 10.1.10.200;
		option routers 10.1.10.1;
		option domain-name-servers 192.168.20.10;
}
```

```
ip helper-address 192.168.20.10
```

```
acl retele_spital {
	127.0.0.1;
	192.168.20.0/24;
	10.1.0.0/16; 10.2.0.0/16; 10.3.0.0/16; 10.4.0.0/16;
};
```

```
options {
	allow-query { retele_spital; };
	recursion yes;
	allow-recursion { retele_spital; };
	forwarders { 8.8.8.8; 1.1.1.1; };
};
```

```
ntp server 10.50.30.10
clock timezone EET 2 0
clock summer-time EEST recurring last Sun Mar 3:00 last Sun Oct 4:00
```

```
service timestamps log datetime msec localtime show-timezone
logging host 10.50.30.10
logging trap informational
```

```
snmp-server community HospitalRO ro
snmp-server location Main-Building
snmp-server contact admin@hospital.local
```

```
aaa new-model
tacacs server AAA-SPITAL
 address ipv4 192.168.30.10
 key SpitalTACACS123!
```

```
aaa group server tacacs+ TACACS-GRP
 server name AAA-SPITAL
aaa authentication login default group TACACS-GRP local
```

```
ansible-galaxy collection install cisco.ios
pip3 install ansible-pylibssh
```

```
vlan 10
 name Zabbix
interface Port-channel1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
```

```
interface range Ethernet1/0 - 1
 channel-group 1 mode active
```

```
interface Ethernet0/0
 switchport port-security maximum 2
 switchport port-security violation shutdown
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
```

```
ip dhcp snooping
ip dhcp snooping vlan 10,20
interface Port-channel1
 ip dhcp snooping trust
```

```
interface Ethernet0/0.5
 encapsulation dot1Q 5
 ip address 192.168.5.2 255.255.255.0
 standby version 2
 standby 5 ip 192.168.5.1
 standby 5 priority 150
 standby 5 preempt
```

```
router ospf 1
 router-id 1.1.1.1
 network 10.0.2.0 0.0.0.3 area 0
 default-information originate
```

```
access-list 1 permit 10.0.0.0 0.255.255.255
access-list 1 permit 192.168.0.0 0.0.255.255
ip nat inside source list 1 interface Ethernet0/2 overload
```

```
crypto isakmp policy 10
 encryption aes 256
 hash sha256
 authentication pre-share
 group 14
```

```
crypto isakmp key SpitalSecurizat123! address 203.0.113.10
```

```
interface Tunnel11
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/2
 tunnel destination 203.0.113.10
 tunnel protection ipsec profile IPSEC-PROF
```

```
access-list OUTSIDE_IN extended permit ospf host 10.0.2.1 any
access-list OUTSIDE_IN extended permit icmp 192.168.0.0 255.255.0.0 any
access-group OUTSIDE_IN in interface outside
```

```
class-map TCP_CLASS
 match access-list TCP_BYPASS
policy-map global_policy
 class TCP_CLASS
  set connection advanced-options tcp-state-bypass
```

```
ip routing
interface Vlan10
 ip address 10.50.10.1 255.255.255.0
interface Vlan20
 ip address 10.50.20.1 255.255.255.0
```

```
acl retele_spital { 127.0.0.1; 192.168.20.0/24;
        10.1.0.0/16; 10.2.0.0/16; 10.3.0.0/16; 10.4.0.0/16; };
options {
        allow-query { retele_spital; };
        allow-recursion { retele_spital; };
        forwarders { 8.8.8.8; 1.1.1.1; };
};
```

```
ip domain name spital.local
crypto key generate rsa modulus 2048
ip ssh version 2
line vty 0 4
 login local
 transport input ssh
```

```
aaa new-model
tacacs server AAA-SPITAL
 address ipv4 192.168.30.10
 key SpitalTACACS123!
aaa authentication login default group TACACS-GRP local
```

```
ntp server 10.50.30.10
clock timezone EET 2 0
service timestamps log datetime msec localtime show-timezone
logging host 10.50.30.10
logging trap informational
```

```
apt install zabbix-server-mysql zabbix-frontend-php \
  zabbix-apache-conf zabbix-sql-scripts zabbix-agent2
```

```
- name: Configurare Banner MOTD (IOS)
  cisco.ios.ios_banner:
        banner: motd
        text: "Acces autorizat - {{ inventory_hostname }}"
        state: present
```

Siteuri Resurse

VPC-uri

ip 10.4.10.10/24 10.4.10.1 192.168.20.10
Show ip
Save

Ip dhcp
ip dns 192.168.20.10

SSH / AAA/NTP

SSH Switches and Routers:

```
configure terminal
ip domain name spital.local
crypto key generate rsa modulus 2048
ip ssh version 2
username gns3 privilege 15 secret gns3
enable secret gns3
line vty 0 4
 login local
 transport input ssh
 Exit
exit
write memory
```

SSH Firewalls:

En
cisco

```
configure terminal
domain-name spital.local
crypto key generate rsa modulus 2048
yes
username gns3 password gns3 privilege 15
aaa authentication ssh console LOCAL
ssh 0.0.0.0 0.0.0.0 inside
write memory
```

AAA Switches and Routers:

```
configure terminal

aaa new-model

tacacs server AAA-SPITAL
 address ipv4 192.168.30.10
 key SpitalTACACS123!
 exit

aaa group server tacacs+ TACACS-GRP
```

```
 server name AAA-SPITAL
 exit

aaa authentication login default group TACACS-GRP local

line vty 0 4
 login authentication default
 exit

end
write memory
```

AAA Firewalls:

```
configure terminal

aaa-server TACACS-GRP protocol tacacs+

aaa-server TACACS-GRP (outside) host 192.168.30.10
 key SpitalTACACS123!
 exit

aaa authentication ssh console TACACS-GRP LOCAL

write memory
```

Verificare:

```
ping 192.168.30.10
show run | include username
show run | include enable secret
test aaa group TACACS-GRP gns3 gns3 legacy
```

NTP Router/Swiches:

configure terminal
ntp server 10.50.30.10
clock timezone EET 2 0
clock summer-time EEST recurring last Sun Mar 3:00 last Sun Oct 4:00
end
write memory

NTP Firewall:

configure terminal
ntp server 10.50.30.10
clock timezone EET 2
end
write memory

Verificare:
show ntp status
show ntp associations
Ping 10.50.30.10

-

Syslog Router/Swiches:

configure terminal
service timestamps log datetime msec localtime show-timezone
logging host 10.50.30.10
logging trap informational
end
write memory

Verifica:
show logging

Syslog Firewall:

configure terminal
logging enable
logging host outside 10.50.30.10

```
logging trap informational
logging timestamp
end
write memory
```

Topologie_Config

Main Building

MB-R1

```
en
conf t
host MB-R1

int e0/3
 description Link_to_MB-FW1_Outside
 ip add 10.0.2.1 255.255.255.252
 no sh

int e0/1
 description Link_to_MB-R2_CrossConnect
 ip add 10.0.4.1 255.255.255.252
 no sh

int e0/2
 description Link_to_ISP
 ip add 203.0.113.2 255.255.255.252
 no sh

exit


!____________________________________________
!Rutare:

ip route 192.168.0.0 255.255.0.0 10.0.2.2 200
ip route 0.0.0.0 0.0.0.0 203.0.113.1
ip route 192.168.40.10 255.255.255.255 10.0.4.2

router ospf 1
router-id 1.1.1.1
network 10.0.2.0 0.0.0.3 area 0
network 10.0.4.0 0.0.0.3 area 0
default-information originate
exit


!____________________________________________
!Nat:


interface Ethernet0/3
 ip nat inside
interface Ethernet0/1
 ip nat inside
interface Ethernet0/2
 ip nat outside
exit
```

```
access-list 1 permit 10.0.0.0 0.255.255.255
access-list 1 permit 192.168.0.0 0.0.255.255
ip nat inside source list 1 interface Ethernet0/2 overload


!_________________________________________
!Tunel:


crypto isakmp policy 10
 encryption aes 256
 hash sha256
 authentication pre-share
 group 14
exit

!Chei dedicate pentru IP-ul public al fiecărui ruter periferic
crypto isakmp key SpitalSecurizat123! address 203.0.113.10
crypto isakmp key SpitalSecurizat123! address 203.0.113.14
crypto isakmp key SpitalSecurizat123! address 203.0.113.18
crypto isakmp key SpitalSecurizat123! address 203.0.113.22
crypto isakmp key SpitalSecurizat123! address 203.0.113.26

!Faza 2 (IPsec)
crypto ipsec transform-set TS-GRE esp-aes esp-sha256-hmac
 mode tunnel
exit

crypto ipsec profile IPSEC-PROF
 set transform-set TS-GRE
exit

!Interfețe Tunel Primare (Spre toate locațiile Spokes)
interface Tunnel11
 description Tunnel_to_EB-R1_Primary
 ip address 10.254.11.1 255.255.255.252
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/2
 tunnel destination 203.0.113.10
 tunnel protection ipsec profile IPSEC-PROF
 no shutdown

interface Tunnel12
 description Tunnel_to_ML-R1_Primary
 ip address 10.254.12.1 255.255.255.252
 ip mtu 1400
```

```
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/2
 tunnel destination 203.0.113.14
 tunnel protection ipsec profile IPSEC-PROF
 no shutdown

interface Tunnel13
 description Tunnel_to_CL-R2_Primary
 ip address 10.254.13.1 255.255.255.252
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/2
 tunnel destination 203.0.113.18
 tunnel protection ipsec profile IPSEC-PROF
 no shutdown

interface Tunnel14
 description Tunnel_to_RS-R1_Primary
 ip address 10.254.14.1 255.255.255.252
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/2
 tunnel destination 203.0.113.22
 tunnel protection ipsec profile IPSEC-PROF
 no shutdown

interface Tunnel15
 description Tunnel_to_RS-R2_Primary
 ip address 10.254.15.1 255.255.255.252
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/2
 tunnel destination 203.0.113.26
 tunnel protection ipsec profile IPSEC-PROF
 no shutdown

! --- Propagare rute prin OSPF Area 0 ---
router ospf 1
 network 10.254.11.0 0.0.0.3 area 0
 network 10.254.12.0 0.0.0.3 area 0
 network 10.254.13.0 0.0.0.3 area 0
 network 10.254.14.0 0.0.0.3 area 0
 network 10.254.15.0 0.0.0.3 area 0
exit
```

!____
!Zabbix:

conf t
snmp-server community HospitalRO ro
snmp-server location Main-Building
snmp-server contact admin@hospital.local
end
wr

MB-R2

```
en
conf t
host MB-R2

int e0/3
 description Link_to_MB-FW2_Outside
 ip add 10.0.3.1 255.255.255.252
 no sh

int e0/1
 description Link_to_MB-R1_CrossConnect
 ip add 10.0.4.2 255.255.255.252
 no sh

int e0/2
 description Link_to_ISP
 ip add 203.0.113.6 255.255.255.252
 no sh

exit


!—-------------------------------------------------------
!Rutare:

conf t
ip route 192.168.0.0 255.255.0.0 10.0.3.2 200
ip route 0.0.0.0 0.0.0.0 203.0.113.5

router ospf 1
router-id 2.2.2.2
network 10.0.3.0 0.0.0.3 area 0
network 10.0.4.0 0.0.0.3 area 0
default-information originate
exit

!____________________________________________
!Nat:


interface Ethernet0/3
 ip nat inside
interface Ethernet0/1
 ip nat inside
interface Ethernet0/2
 ip nat outside
exit
```

```
access-list 1 permit 10.0.0.0 0.255.255.255
access-list 1 permit 192.168.0.0 0.0.255.255
ip nat inside source list 1 interface Ethernet0/2 overload


!__________________________________________
!Tunel:

crypto isakmp policy 10
 encryption aes 256
 hash sha256
 authentication pre-share
 group 14
exit

crypto isakmp key SpitalSecurizat123! address 203.0.113.10
crypto isakmp key SpitalSecurizat123! address 203.0.113.14
crypto isakmp key SpitalSecurizat123! address 203.0.113.18
crypto isakmp key SpitalSecurizat123! address 203.0.113.22
crypto isakmp key SpitalSecurizat123! address 203.0.113.26

! --- Faza 2 (IPsec) ---
crypto ipsec transform-set TS-GRE esp-aes esp-sha256-hmac
 mode tunnel
exit

crypto ipsec profile IPSEC-PROF
 set transform-set TS-GRE
exit

! --- Interfețe Tunel Secundare (Pentru redundanță) ---
interface Tunnel21
 description Tunnel_to_EB-R1_Backup
 ip address 10.254.21.1 255.255.255.252
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/2
 tunnel destination 203.0.113.10
 tunnel protection ipsec profile IPSEC-PROF
 no shutdown

interface Tunnel22
 description Tunnel_to_ML-R1_Backup
 ip address 10.254.22.1 255.255.255.252
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/2
 tunnel destination 203.0.113.14
```

```
 tunnel protection ipsec profile IPSEC-PROF
 no shutdown

interface Tunnel23
 description Tunnel_to_CL-R2_Backup
 ip address 10.254.23.1 255.255.255.252
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/2
 tunnel destination 203.0.113.18
 tunnel protection ipsec profile IPSEC-PROF
 no shutdown

interface Tunnel24
 description Tunnel_to_RS-R1_Backup
 ip address 10.254.24.1 255.255.255.252
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/2
 tunnel destination 203.0.113.22
 tunnel protection ipsec profile IPSEC-PROF
 no shutdown

interface Tunnel25
 description Tunnel_to_RS-R2_Backup
 ip address 10.254.25.1 255.255.255.252
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/2
 tunnel destination 203.0.113.26
 tunnel protection ipsec profile IPSEC-PROF
 no shutdown

! --- Propagare rute prin OSPF Area 0 ---
router ospf 1
 network 10.254.21.0 0.0.0.3 area 0
 network 10.254.22.0 0.0.0.3 area 0
 network 10.254.23.0 0.0.0.3 area 0
 network 10.254.24.0 0.0.0.3 area 0
 network 10.254.25.0 0.0.0.3 area 0
exit
```

!____
!Zabbix:

conf t
snmp-server community HospitalRO ro
snmp-server location Main-Building
snmp-server contact admin@hospital.local
end
wr

MB-R3

```
en
conf t
host MB-R3

int e0/1
no sh

interface e0/1.5
encapsulation dot1q 5
ip address 192.168.5.2 255.255.255.0
standby version 2
standby 5 ip 192.168.5.1
standby 5 priority 150
standby 5 preempt
no shutdown

int e0/1.10
encap dot1q 10
ip add 192.168.10.2 255.255.255.0
standby version 2
standby 10 ip 192.168.10.1
standby 10 prior 150
standby 10 preempt
no sh

int e0/1.20
encap dot1q 20
ip add 192.168.20.2 255.255.255.0
standby version 2
standby 20 ip 192.168.20.1
standby 20 prior 150
standby 20 preempt
no sh

int e0/1.30
encap dot1q 30
ip add 192.168.30.2 255.255.255.0
standby version 2
standby 30 ip 192.168.30.1
no sh

int e0/1.40
encap dot1q 40
ip add 192.168.40.2 255.255.255.0
standby version 2
standby 40 ip 192.168.40.1
no sh
```

```
int e0/0
ip add 10.0.0.1 255.255.255.252
no sh

exit


!—--------------------------------------------------------
!Rutare:

conf t
ip route 0.0.0.0 0.0.0.0 10.0.0.2

router ospf 1
 router-id 3.3.3.3
 network 10.0.0.0 0.0.0.3 area 0
 network 192.168.5.0 0.0.0.255 area 0
 network 192.168.10.0 0.0.0.255 area 0
 network 192.168.20.0 0.0.0.255 area 0
 network 192.168.30.0 0.0.0.255 area 0
 network 192.168.40.0 0.0.0.255 area 0
 passive-interface e0/1.5
 passive-interface e0/1.10
 passive-interface e0/1.20
 passive-interface e0/1.30
 passive-interface e0/1.40
exit
```

MB-R4

```
en
conf t
host MB-R4

int e0/1
no sh

interface e0/1.5
encapsulation dot1q 5
ip address 192.168.5.3 255.255.255.0
standby version 2
standby 5 ip 192.168.5.1
no shutdown


int e0/1.10
encap dot1q 10
ip add 192.168.10.3 255.255.255.0
standby version 2
standby 10 ip 192.168.10.1
no sh

int e0/1.20
encap dot1q 20
ip add 192.168.20.3 255.255.255.0
standby version 2
standby 20 ip 192.168.20.1
no sh

int e0/1.30
encap dot1q 30
ip add 192.168.30.3 255.255.255.0
standby version 2
standby 30 ip 192.168.30.1
standby 30 prior 150
standby 30 preempt
no sh

int e0/1.40
encap dot1q 40
ip add 192.168.40.3 255.255.255.0
standby version 2
standby 40 ip 192.168.40.1
standby 40 prior 150
standby 40 preempt
no sh

int e0/0
```

```
ip add 10.0.1.1 255.255.255.252
no sh
Exit


!—--------------------------------------------------------
!Rutare:

conf t
ip route 0.0.0.0 0.0.0.0 10.0.1.2


router ospf 1
 router-id 4.4.4.4
 network 10.0.1.0 0.0.0.3 area 0
 network 192.168.5.0 0.0.0.255 area 0
 network 192.168.10.0 0.0.0.255 area 0
 network 192.168.20.0 0.0.0.255 area 0
 network 192.168.30.0 0.0.0.255 area 0
 network 192.168.40.0 0.0.0.255 area 0
 passive-interface e0/1.5
 passive-interface e0/1.10
 passive-interface e0/1.20
 passive-interface e0/1.30
 passive-interface e0/1.40
exit



ip route 10.0.0.0 255.255.255.252 192.168.5.2
```

MB-FW1

```
Enable
cisco
configure terminal
hostname MB-FW1

N

! --- 1. Configurare Interfață Interioară (Spre MB-R3) ---
interface GigabitEthernet0/0
 description Link_to_MB-R3
 nameif inside
 security-level 100
 ip address 10.0.0.2 255.255.255.252
 no shutdown
 exit

! --- 2. Configurare Interfață Exterioară (Spre MB-R1) ---
interface GigabitEthernet0/1
 description Link_to_MB-R1
 nameif outside
 security-level 0
 ip address 10.0.2.2 255.255.255.252
 no shutdown
 exit

! --- 3. Activare Inspecție ICMP (Permite trecerea Ping-ului înapoi) ---
policy-map global_policy
 class inspection_default
no  inspect icmp
 Exit
exit


!_______________________________________________
!Rutare:

!---route inside 192.168.0.0 255.255.0.0 10.0.0.1 1
route outside 0.0.0.0 0.0.0.0 10.0.2.1 1

router ospf 1
 router-id 10.0.2.2
 network 10.0.0.0 255.255.255.252 area 0
 network 10.0.2.0 255.255.255.252 area 0
exit
```

```
!________________________________________
!ACL:

access-list OUTSIDE_IN extended permit ospf host 10.0.2.1 any
access-list OUTSIDE_IN extended permit icmp 192.168.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.1.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.2.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.3.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.4.0.0 255.255.0.0 any

access-list OUTSIDE_IN extended permit icmp any 192.168.0.0 255.255.0.0
access-list OUTSIDE_IN extended permit icmp any 10.0.0.0 255.0.0.0

access-group OUTSIDE_IN in interface outside


!—-------------------------------------------------------
!DHCP:

access-list OUTSIDE_IN extended permit udp any host 192.168.20.10 eq bootps
access-list OUTSIDE_IN extended permit udp any host 192.168.20.10 eq domain
access-list OUTSIDE_IN extended permit tcp any host 192.168.20.10 eq domain
write memory



ssh 0.0.0.0 0.0.0.0 outside


configure terminal
access-list TCP_BYPASS extended permit tcp 192.168.0.0 255.255.0.0 any eq 22
access-list TCP_BYPASS extended permit tcp any eq 22 192.168.0.0 255.255.0.0

class-map TCP_CLASS
 match access-list TCP_BYPASS
exit

policy-map global_policy
 class TCP_CLASS
  set connection advanced-options tcp-state-bypass
exit
write memory
```

```
!—-------------------------------------------------------
!AAA:

configure terminal

access-list OUTSIDE_IN extended permit tcp 10.0.0.0 255.0.0.0 host 192.168.30.10 eq 49
access-list OUTSIDE_IN extended permit tcp 192.168.0.0 255.255.0.0 host 192.168.30.10
eq 49

write memory

configure terminal

access-list AAA-BYPASS extended permit tcp any host 192.168.30.10 eq 49
access-list AAA-BYPASS extended permit tcp host 192.168.30.10 eq 49 any

class-map AAA-STATE-BYPASS
 match access-list AAA-BYPASS
 exit

policy-map AAA-POLICY
 class AAA-STATE-BYPASS
  set connection advanced-options tcp-state-bypass
 exit
 exit

service-policy AAA-POLICY interface outside

end
write memory


configure terminal
no service-policy AAA-POLICY interface outside

policy-map global_policy
 class AAA-STATE-BYPASS
  set connection advanced-options tcp-state-bypass
 exit
exit
end
```

MB-FW2

```
Enable

configure terminal
hostname MB-FW2

N


! --- 1. Configurare Interfață Interioară (Spre MB-R4) ---
interface GigabitEthernet0/0
 description Link_to_MB-R4
 nameif inside
 security-level 100
 ip address 10.0.1.2 255.255.255.252
 no shutdown
 exit

! --- 2. Configurare Interfață Exterioară (Spre MB-R2) ---
interface GigabitEthernet0/1
 description Link_to_MB-R2
 nameif outside
 security-level 0
 ip address 10.0.3.2 255.255.255.252
 no shutdown
 exit

! --- 3. Activare Inspecție ICMP (Permite trecerea Ping-ului înapoi) ---
policy-map global_policy
 class inspection_default
no  inspect icmp
 exit
exit

!—-------------------------------------------------------
!Rutare:

!--route inside 192.168.0.0 255.255.0.0 10.0.1.1 1
route outside 0.0.0.0 0.0.0.0 10.0.3.1 1

router ospf 1
 router-id 10.0.1.2
 network 10.0.1.0 255.255.255.252 area 0
 network 10.0.3.0 255.255.255.252 area 0
exit
```

```
!________________________________________
!ACL:

access-list OUTSIDE_IN extended permit ospf host 10.0.3.1 any
access-list OUTSIDE_IN extended permit icmp 192.168.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.1.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.2.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.3.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.4.0.0 255.255.0.0 any

access-list OUTSIDE_IN extended permit icmp any 192.168.0.0 255.255.0.0
access-list OUTSIDE_IN extended permit icmp any 10.0.0.0 255.0.0.0

access-group OUTSIDE_IN in interface outside




!—-------------------------------------------------------
!DHCP:

access-list OUTSIDE_IN extended permit udp any host 192.168.20.10 eq bootps
access-list OUTSIDE_IN extended permit udp any host 192.168.20.10 eq domain
access-list OUTSIDE_IN extended permit tcp any host 192.168.20.10 eq domain
write memory




configure terminal
access-list TCP_BYPASS extended permit tcp 192.168.0.0 255.255.0.0 any eq 22
access-list TCP_BYPASS extended permit tcp any eq 22 192.168.0.0 255.255.0.0

class-map TCP_CLASS
 match access-list TCP_BYPASS
exit

policy-map global_policy
 class TCP_CLASS
  set connection advanced-options tcp-state-bypass
exit
write memory
```

```
!—-----------------------------------------------------
!AAA:

configure terminal

access-list OUTSIDE_IN extended permit tcp 10.0.0.0 255.0.0.0 host 192.168.30.10 eq 49
access-list OUTSIDE_IN extended permit tcp 192.168.0.0 255.255.0.0 host 192.168.30.10
eq 49

write memory


configure terminal

access-list AAA-BYPASS extended permit tcp any host 192.168.30.10 eq 49
access-list AAA-BYPASS extended permit tcp host 192.168.30.10 eq 49 any

class-map AAA-STATE-BYPASS
 match access-list AAA-BYPASS
 exit

policy-map AAA-POLICY
 class AAA-STATE-BYPASS
  set connection advanced-options tcp-state-bypass
 exit
 exit

service-policy AAA-POLICY interface outside

end
write memory

configure terminal
no service-policy AAA-POLICY interface outside

policy-map global_policy
 class AAA-STATE-BYPASS
  set connection advanced-options tcp-state-bypass
 exit
exit
end



configure terminal
access-list OUTSIDE_IN extended permit tcp host 192.168.30.10 eq 49 10.0.0.0 255.0.0.0
access-list OUTSIDE_IN extended permit tcp host 192.168.30.10 eq 49 192.168.0.0
255.255.0.0
```

```
write memory

configure terminal
access-list OUTSIDE_IN extended permit udp host 192.168.10.10 any eq 161
access-list OUTSIDE_IN extended permit udp any eq 161 host 192.168.10.10
write memory
```

MB-SW1

```
En
Conf t
Host MB-SW1

vlan 5
name Management
vlan 10
name Zabbix
vlan 20
name DNS-DHCP
vlan 30
name AAA
vlan 40
name Ansible
vlan 99
name Native
exit


interface e0/1
description Uplink_to_MB-R3
!Duplex full
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
no shutdown
exit


interface range e1/0-1
description Po1_to_MB-SW3
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
channel-group 1 mode active
no shutdown
exit

interface Port-channel1
description Trunk_to_MB-SW3
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
no shutdown
exit
```

```
interface range e1/2-3
description Po2_to_MB-SW4
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
channel-group 2 mode active
no shutdown
Exit

interface Port-channel2
description Trunk_to_MB-SW4
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
no shutdown
Exit


interface range e0/0, e0/2
description Po3_to_MB-SW2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
channel-group 3 mode active
no shutdown
Exit

interface Port-channel3
description Trunk_to_MB-SW2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
no shutdown
exit


interface Vlan5
description Management
ip address 192.168.5.11 255.255.255.0
no shutdown
Exit
```

ip default-gateway 192.168.5.1

No ip routing

spanning-tree mode rapid-pvst
spanning-tree vlan 10,20 priority 4096
spanning-tree vlan 5,30,40,99 priority 8192

!no ip igmp snooping (Switch-TEST)

MB-SW2

```
En
Conf t
Host MB-SW2

vlan 5
name Management
vlan 10
name Zabbix
vlan 20
name DNS-DHCP
vlan 30
name AAA
vlan 40
name Ansible
vlan 99
name Native
exit


interface e0/1
description Uplink_to_MB-R4
!Duplex full
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
no shutdown
exit


interface range e1/0-1
description Po1_to_MB-SW4
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
channel-group 1 mode active
no shutdown
exit

interface Port-channel1
description Trunk_to_MB-SW4
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
no shutdown
exit
```

```
interface range e1/2-3
description Po2_to_MB-SW3
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
channel-group 2 mode active
no shutdown
Exit

interface Port-channel2
description Trunk_to_MB-SW3
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
no shutdown
Exit


interface range e0/0, e0/2
description Po3_to_MB-SW1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
channel-group 3 mode active
no shutdown
Exit

interface Port-channel3
description Trunk_to_MB-SW1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
no shutdown
exit


interface Vlan5
description Management
ip address 192.168.5.12 255.255.255.0
no shutdown
Exit
```

```
ip default-gateway 192.168.5.1

No ip routing

spanning-tree mode rapid-pvst
spanning-tree vlan 30,40 priority 4096
spanning-tree vlan 5,10,20,99 priority 8192

!no ip igmp snooping (Switch-TEST)
```

MB-SW3

```
En
Conf t
Host MB-SW3

vlan 5
name Management
vlan 10
name Zabbix
vlan 20
name DNS-DHCP
vlan 30
name AAA
vlan 40
name Ansible
vlan 99
name Native
exit


interface e0/0
description Access_Zabbix_VLAN10
switchport mode access
 switchport access vlan 10
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown

interface e0/1
description Access_DNS-DHCP_VLAN20
switchport mode access
switchport access vlan 20
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown


interface range e1/0-1
description Po1_to_MB-SW1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
channel-group 1 mode active
no shutdown
exit

interface Port-channel1
description Trunk_to_MB-SW1
```

```
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
no shutdown
exit


interface range e1/2-3
description Po2_to_MB-SW2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
channel-group 2 mode active
no shutdown
Exit

interface Port-channel2
description Trunk_to_MB-SW2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
no shutdown
exit


interface Vlan5
description Management
ip address 192.168.5.13 255.255.255.0
no shutdown
Exit

ip default-gateway 192.168.5.1

No ip routing

spanning-tree mode rapid-pvst

!no ip igmp snooping (Switch-TEST)
```

!___________________________________

!Port security:

```
configure terminal
ip dhcp snooping
no ip dhcp snooping information option
ip dhcp snooping vlan 10,20

interface range Ethernet0/0 - 1
 switchport port-security
 switchport port-security maximum 2
 switchport port-security violation shutdown
 switchport port-security mac-address sticky
 ip dhcp snooping limit rate 15

interface Port-channel1
 ip dhcp snooping trust
interface Port-channel2
 ip dhcp snooping trust
exit
write memory
```

MB-SW4

```
En
Conf t
Host MB-SW4

vlan 5
name Management
vlan 10
name Zabbix
vlan 20
name DNS-DHCP
vlan 30
name AAA
vlan 40
name Ansible
vlan 99
name Native
exit


interface e0/0
description Access_AAA_VLAN30
switchport mode access
 switchport access vlan 30
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown
exit

interface e0/1
description Access_Ansible_VLAN40
switchport mode access
switchport access vlan 40
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown
exit


interface range e1/0-1
description Po1_to_MB-SW2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
channel-group 1 mode active
no shutdown
exit
```

```
interface Port-channel1
description Trunk_to_MB-SW2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
no shutdown
exit


interface range e1/2-3
description Po2_to_MB-SW1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
channel-group 2 mode active
no shutdown
Exit

interface Port-channel2
description Trunk_to_MB-SW1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 5,10,20,30,40,99
no shutdown
exit


interface Vlan5
description Management
ip address 192.168.5.14 255.255.255.0
no shutdown
Exit

ip default-gateway 192.168.5.1

No ip routing

spanning-tree mode rapid-pvst

!no ip igmp snooping (Switch-TEST)
```

!_____________________________________
!Port security:

```
configure terminal
ip dhcp snooping
no ip dhcp snooping information option
ip dhcp snooping vlan 30,40

interface range Ethernet0/0 - 1
 switchport port-security
 switchport port-security maximum 2
 switchport port-security violation shutdown
 switchport port-security mac-address sticky
 ip dhcp snooping limit rate 15

interface Port-channel1
 ip dhcp snooping trust
interface Port-channel2
 ip dhcp snooping trust
exit
write memory
```

Zabbix

Main ZABBIX:

```
#!/bin/bash
# ==============================================================
# Zabbix 7.0 LTS install on Ubuntu 22.04 (MariaDB + Apache)
# Change ubuntu22.04 to ubuntu20.04 / ubuntu24.04 if needed.
# Replace ParolaTaPuternica and the IPs/SNMP creds before use.
# ==============================================================

# 1. Update the system
sudo apt update && sudo apt upgrade -y

# 2. Install the database (MariaDB)
sudo apt install mariadb-server -y
sudo systemctl enable --now mariadb
sudo mysql_secure_installation   # interactive: set root pw, answer Y to hardening

# 3. Add the Zabbix repository
wget
https://repo.zabbix.com/zabbix/7.0/ubuntu/pool/main/z/zabbix-release/zabbix-release_latest+
ubuntu22.04_all.deb
sudo dpkg -i zabbix-release_latest+ubuntu22.04_all.deb
sudo apt update

# 4. Install Zabbix server, frontend, and agent
sudo apt install zabbix-server-mysql zabbix-frontend-php zabbix-apache-conf
zabbix-sql-scripts zabbix-agent2 -y

# 5. Create the Zabbix database and user
#    (runs the SQL non-interactively; you'll be prompted for the MariaDB root password)
sudo mysql -uroot -p <<'SQL'
CREATE DATABASE zabbix CHARACTER SET utf8mb4 COLLATE utf8mb4_bin;
CREATE USER 'zabbix'@'localhost' IDENTIFIED BY 'ParolaTaPuternica';
GRANT ALL PRIVILEGES ON zabbix.* TO 'zabbix'@'localhost';
SET GLOBAL log_bin_trust_function_creators = 1;
SQL

# 6. Import the initial schema (enter the zabbix user password when prompted)
sudo zcat /usr/share/zabbix-sql-scripts/mysql/server.sql.gz | mysql
--default-character-set=utf8mb4 -uzabbix -p zabbix
#    Disable the temporary setting again:
sudo mysql -uroot -p -e "SET GLOBAL log_bin_trust_function_creators = 0;"

# 7. Set the DB password in the server config
#    Edit /etc/zabbix/zabbix_server.conf and set: DBPassword=ParolaTaPuternica
sudo sed -i 's/^# DBPassword=.*/DBPassword=ParolaTaPuternica/'
/etc/zabbix/zabbix_server.conf
```

```
# 8. Set the PHP timezone for the frontend
#    Edit /etc/zabbix/apache.conf and uncomment/set the timezone line:
sudo sed -i 's|^.*php_value date.timezone.*|        php_value date.timezone
Europe/Bucharest|' /etc/zabbix/apache.conf

# 9. Start and enable all services
sudo systemctl restart zabbix-server zabbix-agent2 apache2
sudo systemctl enable zabbix-server zabbix-agent2 apache2
#    Verify the server is healthy:
sudo systemctl status zabbix-server
#    (optional) watch the log for DB errors: sudo tail -f /var/log/zabbix/zabbix_server.log

# 10. Open the firewall ports
sudo ufw allow 80/tcp        # web frontend
sudo ufw allow 10051/tcp     # server (active agents/traps in)
sudo ufw allow 10050/tcp     # agent (server polls hosts)
sudo ufw reload

# 11. Finish in the browser (UI step, no command)
#     Open http://<server-ip>/zabbix , complete the wizard,
#     log in with Admin / zabbix , then change the password.

# 12. Verify SNMP reachability before adding devices
sudo apt install snmp -y
#     SNMPv2c test:
snmpwalk -v2c -c YOUR_COMMUNITY 192.168.10.1
#     SNMPv3 test (recommended on ASAv/IOS):
snmpwalk -v3 -l authPriv -u zbxuser -a SHA -A AuthPass -x AES -X PrivPass 192.168.10.1

# ============================================================
# 13. Run ONLY on your OTHER Linux hosts (DNS/DHCP, Ansible),
#     NOT on the Zabbix server itself.
# ============================================================
# wget
https://repo.zabbix.com/zabbix/7.0/ubuntu/pool/main/z/zabbix-release/zabbix-release_latest+
ubuntu22.04_all.deb
# sudo dpkg -i zabbix-release_latest+ubuntu22.04_all.deb
# sudo apt update
# sudo apt install zabbix-agent2 -y
# Edit /etc/zabbix/zabbix_agent2.conf -> set:
#   Server=192.168.10.5
#   ServerActive=192.168.10.5
#   Hostname=DNS-DHCP-Server
# sudo systemctl restart zabbix-agent2
# sudo systemctl enable zabbix-agent2
```

Extra:

Two cautions on the `sed` lines (steps 7 and 8): they assume the default commented format shipped by the package. After running, quickly confirm they took effect:

```
grep '^DBPassword=' /etc/zabbix/zabbix_server.conf
grep 'date.timezone' /etc/zabbix/apache.conf
```

If either comes back empty or wrong, just open the file with `sudo nano` and set the line by hand — it's the more reliable route if the package's default formatting differs from what the `sed` expects.

Step by step applied:

Adaugat ip initial+default gateway:

ip a (Vedem ca nu avem ip)

sudo nano /etc/netplan/01-network-manager-all.yaml

```
network:
  version: 2
  renderer: NetworkManager
  ethernets:
    ens3:  # <-- Replace with your interface name from step 2
      dhcp4: no
      addresses:
        -  192.168.10.10/24
      routes:
        - to: default
          via: 192.168.10.1
      nameservers:
        addresses: [8.8.8.8, 127.0.0.1]
```

```
sudo netplan apply

sudo apt update
sudo dpkg --configure -a
```

wget
https://repo.zabbix.com/zabbix/6.0/ubuntu/pool/main/z/zabbix-release/zabbix-release_6.0-4+ubuntu22.04_all.deb

sudo dpkg -i zabbix-release_6.0-4+ubuntu22.04_all.deb

sudo apt update

apt install zabbix-server-mysql zabbix-frontend-php zabbix-apache-conf zabbix-sql-scripts zabbix-agent mariadb-server -y

apt install mariadb-server mariadb-client -y
systemctl enable mariadb
systemctl start mariadb
systemctl status mariadb

mysql -u root

CREATE DATABASE zabbix CHARACTER SET utf8mb4 COLLATE utf8mb4_bin;
CREATE USER 'zabbix'@'localhost' IDENTIFIED BY 'password';
GRANT ALL PRIVILEGES ON zabbix.* TO 'zabbix'@'localhost';
SET GLOBAL log_bin_trust_function_creators = 1;
QUIT;

zcat /usr/share/zabbix-sql-scripts/mysql/server.sql.gz | mysql
--default-character-set=utf8mb4 -uzabbix -p zabbix

sudo nano /etc/zabbix/zabbix_server.conf
ctrl+w -> search DBPassword= and uncomment

systemctl restart zabbix-server zabbix-agent apache2
systemctl enable zabbix-server zabbix-agent apache2

Host Groups:

Configuration → Host groups → Create host group:

- Hospital/Main Building
- Hospital/Emergency Building
- Hospital/Medical Lab
- Hospital/Clinic
- Hospital/Remote Site

Add Host:

conf t
snmp-server community HospitalRO ro
snmp-server location Main-Building
snmp-server contact admin@hospital.local
end
Wr

snmpwalk -v2c -c HospitalRO <ip device> .1.3.6.1.2.1.1.1.0

SNMP răspunde. Acum adaugă-l în Zabbix web UI:

1. Click Create host (butonul din dreapta-sus)

2. Tab Host:

- Host name: `MB-R1`
- Groups: click Select → alegi grupul tău (ex. Hospital/Main Building dacă l-ai creat, altfel Templates/Network devices)
- Interfaces: click Add → alegi SNMP
  - IP address: `10.0.2.1`
  - DNS name: lasă gol
  - Connect to: IP
  - Port: `161`

3. Tab Templates:

- În câmpul Link new templates scrii `Cisco IOS` → din dropdown alegi Template Net Cisco IOS SNMP

4. Tab Macros:

- Click Add
- Macro: `{$SNMP_COMMUNITY}`
- Value: `HospitalRO`

5. Click Add (butonul albastru de jos)

Create email:

1. Configurezi email media type

Administration → Media types → click pe **Email** (există deja):

- SMTP server: `smtp.mail.yahoo.com`
- SMTP server port: `587`
- SMTP helo: `yahoo.com`
- SMTP email: `adresa-ta@yahoo.com`
- Security: **STARTTLS**
- Authentication: **Username and password**
- Username: `adresa-ta@yahoo.com`
- Password: trebuie **App Password**

Pentru App Password: mergi în Google Account → Security → 2-Step Verification → App passwords → generezi unul pentru "Mail".

Apoi click **Update**.

2. Adaugi email la userul Admin

Administration → Users → **Admin** → tab **Media** → click **Add**:

- Type: **Email**
- Send to: `adresa-ta@gmail.com`
- When active: `1-7,00:00-24:00`
- Severity: bifează **Warning, Average, High, Disaster**

Click Add → apoi **Update** pe user.

3. Creezi Action-ul

Configuration → Actions → **Trigger actions** → **Create action**

Tab **Action**:

- Name: `Interface Down Alert`
- Conditions → Add: Type **Trigger severity** → **is greater than or equals** → **Warning**

Tab **Operations** → Add:

- Send to users: **Admin**
- Send only to: **Email**

Click Add → apoi **Add** pe action.

DNS-DHCP

Parola: osboxes.org
Adaugat ip initial+default gateway:
ip a (Vedem ca nu avem ip)
sudo nano /etc/netplan/01-network-manager-all.yaml

```
network:
  version: 2
  renderer: NetworkManager
  ethernets:
    ens3:  # <-- Replace with your interface name from step 2
      dhcp4: no
      addresses:
        - 192.168.20.10/24
      routes:
        - to: default
          via: 192.168.20.1
      nameservers:
        addresses: [8.8.8.8, 127.0.0.1]
```

sudo netplan apply

Instalat DHCP:

Sudo apt update
sudo apt install isc-dhcp-server bind9 -y

FILE: /etc/dhcp/dhcpd.conf

```
# ----------------------------------------------------------------
# CONFIGURARE GLOBALĂ DHCP - COMPLEX SPITALICESC COMPLETE
# ----------------------------------------------------------------

ddns-update-style none;
authoritative;

default-lease-time 86400;
max-lease-time 172800;

# 1. SCOPE LOCAL: Zona Area 0 (Clădirea Principală - VLAN 20)
subnet 192.168.20.0 netmask 255.255.255.0 {
}

# ================================================================
# 2. SCOPE-URI REMOTE: ZONA AREA 1 (Emergency Building)
# ================================================================
```

```
}

# VLAN 10 - ICU
subnet 10.1.10.0 netmask 255.255.255.0 {
   range 10.1.10.50 10.1.10.200;
   option routers 10.1.10.1;
   option domain-name-servers 192.168.20.10;
}

# VLAN 20 - Radiology
subnet 10.1.20.0 netmask 255.255.255.0 {
   range 10.1.20.50 10.1.20.200;
   option routers 10.1.20.1;
   option domain-name-servers 192.168.20.10;
}

# VLAN 30 - Triage
subnet 10.1.30.0 netmask 255.255.255.0 {
   range 10.1.30.50 10.1.30.200;
   option routers 10.1.30.1;
   option domain-name-servers 192.168.20.10;
}

# VLAN 40 - Local-Server
subnet 10.1.40.0 netmask 255.255.255.0 {
   range 10.1.40.50 10.1.40.200;
   option routers 10.1.40.1;
   option domain-name-servers 192.168.20.10;
}

# ==========================================================
# 3. SCOPE-URI REMOTE: ZONA AREA 2 (Medical Lab)
# ==========================================================

# VLAN 10 - LabPC
subnet 10.2.10.0 netmask 255.255.255.0 {
   range 10.2.10.50 10.2.10.200;
   option routers 10.2.10.1;
   option domain-name-servers 192.168.20.10;
}

# VLAN 20 - PCRMachine
subnet 10.2.20.0 netmask 255.255.255.0 {
   range 10.2.20.50 10.2.20.200;
   option routers 10.2.20.1;
   option domain-name-servers 192.168.20.10;
}

# VLAN 30 - Analyzer
subnet 10.2.30.0 netmask 255.255.255.0 {
```

```
    range 10.2.30.50 10.2.30.200;
    option routers 10.2.30.1;
    option domain-name-servers 192.168.20.10;
}

# VLAN 40 - Local-Database
subnet 10.2.40.0 netmask 255.255.255.0 {
    range 10.2.40.50 10.2.40.200;
    option routers 10.2.40.1;
    option domain-name-servers 192.168.20.10;
}

# ======================================================================
# 4. SCOPE-URI REMOTE: ZONA AREA 3 (Clinic)
# ======================================================================

# VLAN 10 - DoctorPC
subnet 10.3.10.0 netmask 255.255.255.0 {
    range 10.3.10.50 10.3.10.200;
    option routers 10.3.10.1;
    option domain-name-servers 192.168.20.10;
}

# VLAN 20 - ReceptionPC
subnet 10.3.20.0 netmask 255.255.255.0 {
    range 10.3.20.50 10.3.20.200;
    option routers 10.3.20.1;
    option domain-name-servers 192.168.20.10;
}

# VLAN 30 - NurseTerminal
subnet 10.3.30.0 netmask 255.255.255.0 {
    range 10.3.30.50 10.3.30.200;
    option routers 10.3.30.1;
    option domain-name-servers 192.168.20.10;
}

# VLAN 40 - Local-Server1
subnet 10.3.40.0 netmask 255.255.255.0 {
    range 10.3.40.50 10.3.40.200;
    option routers 10.3.40.1;
    option domain-name-servers 192.168.20.10;
}
```

Instalare DNS:

sudo nano /etc/bind/named.conf.options

```
// 1. Definim rețelele din spital care au voie să ceară DNS de la acest server
acl retele_spital {
    127.0.0.1;
    192.168.20.0/24;  // Area 0 - Clădirea Principală (Servere)
    10.1.0.0/16;      // Area 1 - Urgențe (Toate VLAN-urile)
    10.2.0.0/16;      // Area 2 - Laborator
    10.3.0.0/16;      // Area 3 - Clinică
    10.4.0.0/16;      // Area 4 - Remote Site
};

options {
    directory "/var/cache/bind";

    // Securitate: Permite interogări DOAR de la calculatoarele spitalului
    allow-query { retele_spital; };

    // Permite rezolvarea recursivă a numelor (căutarea pe internet)
    recursion yes;
    allow-recursion { retele_spital; };

    // 2. FORWARDERS: Serverele globale către care mașina trimite cererile
    // Când un PC cere "google.com", serverul tău nu îl știe local, așa că întreabă Google
public.
    forwarders {
        8.8.8.8;
        1.1.1.1;
    };

    // Ascultă pe toate interfețele pe portul standard 53
    listen-on port 53 { any; };
    listen-on-v6 { none; }; # Dezactivăm IPv6 deoarece nu îl folosim în rețea

    dnssec-validation auto;
};
```

---

sudo nano /etc/default/named
OPTIONS="-u bind -4"

AAA

ip a   # vezi numele interfeței (ens3)
sudo nano /etc/netplan/01-network-manager-all.yaml

```
network:
  version: 2
  renderer: NetworkManager
  ethernets:
    ens3:
      dhcp4: no
      addresses:
        - 192.168.30.10/24
      routes:
        - to: default
          via: 192.168.30.1
      nameservers:
        addresses: [192.168.20.10, 8.8.8.8]
```

---

sudo netplan apply

sudo apt update
sudo add-apt-repository universe
sudo apt update
sudo apt install python3-pip -y
sudo apt install software-properties-common -y

cd /tmp

wget
"http://archive.ubuntu.com/ubuntu/pool/universe/t/tacacs+/libtacacs+1_4.0.4.27a-3_amd64.deb"

wget
"http://archive.ubuntu.com/ubuntu/pool/universe/t/tacacs+/tacacs+_4.0.4.27a-3_amd64.deb"

sudo apt install ./libtacacs+1_4.0.4.27a-3_amd64.deb ./tacacs+_4.0.4.27a-3_amd64.deb -y
sudo apt install -f -y

echo "deb http://archive.ubuntu.com/ubuntu bionic universe main" | sudo tee /etc/apt/sources.list.d/bionic-tacacs.list
sudo apt update

sudo apt install python2 -y

sudo dpkg -i --ignore-depends=python libtacacs+1_4.0.4.27a-3_amd64.deb tacacs+_4.0.4.27a-3_amd64.deb

sudo ln -s /usr/bin/python2 /usr/bin/python
sudo apt update
sudo apt install python2 -y

sudo rm /etc/apt/sources.list.d/bionic-tacacs.list
sudo apt update

sudo nano /etc/tacacs+/tac_plus.conf

```
key = SpitalTACACS123!

group = admins {
    default service = permit
    service = exec {
        priv-lvl = 15
    }
}

user = gns3 {
    member = admins
    login = cleartext gns3
}
```

---

sudo /etc/init.d/tacacs_plus restart
sudo /etc/init.d/tacacs_plus status

sudo ss -tlnp | grep :49   (liste …..0.0.0.0:49)

Ansible

Parola: osboxes.org

rm ~/.ssh/known_hosts (Sters key retinute)
rm -rf ~/.ansible/pc/*
rm -rf ~/.ansible/cp/*

Adaugat ip initial+default gateway:
ip a (Vedem ca nu avem ip)
sudo nano /etc/netplan/01-network-manager-all.yaml

```
network:
  version: 2
  renderer: NetworkManager
  ethernets:
    ens3:  # <-- Replace with your interface name from step 2
      dhcp4: no
      addresses:
        - 192.168.40.10/24
      routes:
        - to: default
          via: 192.168.40.1
      nameservers:
        addresses: [8.8.8.8, 127.0.0.1]
```

sudo netplan apply

Install Ansible:

sudo apt update

sudo apt install software-properties-common -y
sudo add-apt-repository --yes --update ppa:ansible/ansible
sudo apt install ansible -y
ansible-galaxy collection install cisco.ios
ansible --version

sudo apt install python3-pip -y
pip3 install ansible-pylibssh

Prepare Ansible:

nano ansible.cfg

```
[defaults]
inventory = hosts
```

host_key_checking = False
retry_files_enabled = True

Inventar: nano hosts

```
# -----------------------------------------
# AREA 0 (Main Building / Core Backbone)
# -----------------------------------------
[area0_routers]
MB-R1 ansible_host=10.0.2.1
MB-R2 ansible_host=10.0.3.1
MB-R3 ansible_host=192.168.5.2
MB-R4 ansible_host=192.168.5.3

[area0_switches]
MB-SW1 ansible_host=192.168.5.11
MB-SW2 ansible_host=192.168.5.12
MB-SW3 ansible_host=192.168.5.13
MB-SW4 ansible_host=192.168.5.14

[area0_firewalls]
MB-FW1 ansible_host=10.0.2.2
MB-FW2 ansible_host=10.0.1.2

[area0:children]
area0_routers
area0_switches
area0_firewalls


# -----------------------------------------
# AREA 1 (Emergency Building - Spoke)
# -----------------------------------------
[area1_routers]
EB-R1 ansible_host=10.1.102.1

[area1_switches]
EB-SW-CORE ansible_host=10.1.5.1
EB-SW1 ansible_host=10.1.5.11
EB-SW2 ansible_host=10.1.5.12

[area1_firewalls]
EB-FW1 ansible_host=10.1.102.2

[area1:children]
area1_routers
```

```
area1_switches
area1_firewalls


# -----------------------------------------
# AREA 2 (Medical Lab - Spoke)
# -----------------------------------------
[area2_routers]
ML-R1 ansible_host=10.2.102.1

[area2_switches]
ML-SW-CORE ansible_host=10.2.5.1
ML-SW1 ansible_host=10.2.5.11
ML-SW2 ansible_host=10.2.5.12

[area2_firewalls]
ML-FW1 ansible_host=10.2.102.2

[area2:children]
area2_routers
area2_switches
area2_firewalls


# -----------------------------------------
# AREA 3 (Clinic - Spoke)
# -----------------------------------------
[area3_routers]
CL-R2 ansible_host=10.3.102.1

[area3_switches]
CL-SW-CORE ansible_host=10.3.5.1
CL-SW1 ansible_host=10.3.5.11
CL-SW2 ansible_host=10.3.5.12

[area3_firewalls]
CL-FW1 ansible_host=10.3.102.2

[area3:children]
area3_routers
area3_switches
area3_firewalls


# -----------------------------------------
# AREA 4 (Remote Site - Spoke)
# -----------------------------------------
[area4_routers]
```

```
RS-R1 ansible_host=10.4.102.1
RS-R2 ansible_host=10.4.103.1

[area4_switches]
RS-SW1 ansible_host=10.4.5.2
RS-SW2 ansible_host=10.4.5.3
RS-SW3 ansible_host=10.4.5.13
RS-SW4 ansible_host=10.4.5.14

[area4_firewalls]
RS-FW1 ansible_host=10.4.102.2
RS-FW2 ansible_host=10.4.103.2

[area4:children]
area4_routers
area4_switches
area4_firewalls


# ----------------------------------------
# GRUPURI SPECIALE (Doar Firewall-uri / ASA)
# ----------------------------------------
[all_firewalls:children]
area0_firewalls
area1_firewalls
area2_firewalls
area3_firewalls
area4_firewalls

[all_firewalls:vars]
ansible_connection=network_cli
ansible_network_os=cisco.asa.asa
ansible_user=gns3
ansible_password=gns3
ansible_become=yes
ansible_become_method=enable
ansible_become_password=cisco
ansible_network_cli_ssh_type=paramiko
ansible_ssh_common_args='-o HostKeyAlgorithms=+ssh-rsa -o
PubkeyAcceptedKeyTypes=+ssh-rsa -o Ciphers=aes256-cbc'



# ----------------------------------------
# GRUPURI SPECIALE (Doar Routere și Switch-uri / IOS)
# ----------------------------------------
[all_ios:children]
area0_routers
```

```
area0_switches
area1_routers
area1_switches
area2_routers
area2_switches
area3_routers
area3_switches
area4_routers
area4_switches

[all_ios:vars]
ansible_user=gns3
ansible_password=gns3
ansible_connection=network_cli
ansible_network_os=cisco.ios.ios
ansible_become=yes
ansible_become_method=enable
ansible_become_password=gns3
ansible_ssh_common_args='-o HostKeyAlgorithms=+ssh-rsa -o Ciphers=aes256-cbc -o PubkeyAuthentication=no'
```

Playbooks:

```
nano set_banners.yml
ansible-playbook set_banners.yml
```

```
---
# =====================================================================
# 1. Aplicare Banner pe Routere și Switch-uri (Cisco IOS)
# =====================================================================
- name: Configurare Banner MOTD pe echipamentele infrastructurii (IOS)
  hosts: all_ios
  gather_facts: no

  tasks:
    - name: Setare banner login (MOTD)
      cisco.ios.ios_banner:
        banner: motd
        text: |
          ********************************************************
          * ACCES RESTRICTIONAT - INFRASTRUCTURA SPITAL          *
          * Echipament: {{ inventory_hostname }}                 *
          * Daca nu aveti autorizatie, deconectati-va imediat!  *
          ********************************************************
        state: present

# =====================================================================
# 2. Aplicare Banner pe Firewall-uri (Cisco ASA)
```

```
# ======================================================================
- name: Configurare Banner MOTD pe Firewall-urile perimetrale (ASA)
  hosts: all_firewalls
  gather_facts: no

  tasks:
    - name: Setare banner login (MOTD)
      cisco.asa.asa_banner:
      Lines:
            - 'banner motd ********************************************************'
            - 'banner motd * AVERTISMENT DE SECURITATE - FIREWALL ENTERPRISE *'
            - 'banner motd * Echipament: {{ inventory_hostname }} *'
            - 'banner motd * Toate actiunile sunt inregistrate si monitorizate! *'
            - 'banner motd ********************************************************'
```

Emergency Building

EB-R1

```
en
conf t
host EB-R1

int e0/0
 description Link_to_EB-FW1_Outside
 ip add 10.1.102.1 255.255.255.252
 no sh

int e0/1
 description Link_to_ISP
 ip add 203.0.113.10 255.255.255.252
 no sh

exit


!___________________________________________
!Rutare:

ip route 0.0.0.0 0.0.0.0 203.0.113.9

router ospf 1
 router-id 11.1.1.1
 network 10.1.102.0 0.0.0.3 area 1
exit

!___________________________________________
!Nat:

interface Ethernet0/0
 ip nat inside
interface Ethernet0/1
 ip nat outside
exit

access-list 1 permit 10.1.0.0 0.0.255.255
ip nat inside source list 1 interface Ethernet0/1 overload

!___________________________________________
!Tunel:


crypto isakmp policy 10
 encryption aes 256
 hash sha256
 authentication pre-share
 group 14
```

```
exit

crypto isakmp key SpitalSecurizat123! address 203.0.113.2
crypto isakmp key SpitalSecurizat123! address 203.0.113.6

crypto ipsec transform-set TS-GRE esp-aes esp-sha256-hmac
 mode tunnel
exit

crypto ipsec profile IPSEC-PROF
 set transform-set TS-GRE
exit

! Tunel spre MB-R1 (Calea Primară)
interface Tunnel11
 description Tunnel_to_MB-R1_Primary
 ip address 10.254.11.2 255.255.255.252
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/1
 tunnel destination 203.0.113.2
 tunnel protection ipsec profile IPSEC-PROF
 no shutdown

! Tunel spre MB-R2 (Calea Secundară)
interface Tunnel21
 description Tunnel_to_MB-R2_Backup
 ip address 10.254.21.2 255.255.255.252
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/1
 tunnel destination 203.0.113.6
 tunnel protection ipsec profile IPSEC-PROF
 no shutdown
exit

router ospf 1
 network 10.254.11.0 0.0.0.3 area 0
 network 10.254.21.0 0.0.0.3 area 0
exit
```

```
!_______________________________________________
!AAA:

configure terminal
interface Tunnel21
 ip ospf cost 5000
end
write memory
```

EB-FW1

```
Enable
cisco
configure terminal
hostname EB-FW1

! --- 1. Configurare Interfață Interioară (Spre EB-SW-CORE) ---
interface GigabitEthernet0/1
 description Link_to_EB-SW-CORE
 nameif inside
 security-level 100
 ip address 10.1.100.2 255.255.255.252
 no shutdown
 exit

! --- 2. Configurare Interfață Exterioară (Spre EB-R1) ---
interface GigabitEthernet0/0
 description Link_to_EB-R1
 nameif outside
 security-level 0
 ip address 10.1.102.2 255.255.255.252
 no shutdown
 exit

! --- 3. Activare Inspecție ICMP (Modular Policy Framework) ---
policy-map global_policy
 class inspection_default
  inspect icmp
 exit
exit

! --- 4. Rutare Implicită (Spre Internet via EB-R1) ---
route outside 0.0.0.0 0.0.0.0 10.1.102.1 1

! --- 5. Motorul de Rutare Dinamică OSPF Area 1 ---
router ospf 1
 router-id 10.1.102.2
 network 10.1.100.0 255.255.255.252 area 1
 network 10.1.102.0 255.255.255.252 area 1
 default-information originate always
exit
```

```
!________________________________________
!ACL:


access-list OUTSIDE_IN extended permit ospf host 10.1.102.1 any
access-list OUTSIDE_IN extended permit icmp 192.168.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.1.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.2.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.3.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.4.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit tcp 192.168.0.0 255.255.0.0 any eq 22
access-group OUTSIDE_IN in interface outside


ssh 0.0.0.0 0.0.0.0 outside
```

EB-SW-CORE

```
en
conf t
host EB-SW-CORE

! --- 1. Activare Rutare L3 și Rapid Spanning Tree ---
ip routing
spanning-tree mode rapid-pvst
spanning-tree vlan 5,10,20,30,40 priority 4096

! --- 2. Definire Bază de Date VLAN-uri ---
vlan 5
 name Management
vlan 10
 name ICU
vlan 20
 name Radiology
vlan 30
 name Triage
vlan 40
 name Local-Server
vlan 99
 name Native
exit

! --- 3. Legătură Downstream către EB-SW1 (Port-channel 1) ---
interface range e1/0-1
 description Link_to_EB-SW1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 channel-group 1 mode active
 no shutdown
 exit

interface Port-channel1
 description Trunk_to_EB-SW1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 no shutdown
 exit

! --- 4. Legătură Downstream către EB-SW2 (Port-channel 2) ---
interface range e1/2-3
 description Link_to_EB-SW2
 switchport trunk encapsulation dot1q
```

```
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 channel-group 2 mode active
 no shutdown
 exit

interface Port-channel2
 description Trunk_to_EB-SW2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 no shutdown
 exit

! --- 5. Configurare SVI-uri (Default Gateways pentru VLAN-uri) ---
interface Vlan5
 description Gateway_Management
 ip address 10.1.5.1 255.255.255.0
 no shutdown

interface Vlan10
 description Gateway_ICU
 ip address 10.1.10.1 255.255.255.0
 no shutdown

interface Vlan20
 description Gateway_Radiology
 ip address 10.1.20.1 255.255.255.0
 no shutdown

interface Vlan30
 description Gateway_Triage
 ip address 10.1.30.1 255.255.255.0
 no shutdown

interface Vlan40
 description Gateway_Local-Server
 ip address 10.1.40.1 255.255.255.0
 no shutdown

! --- 6. Pregătire Port către Firewall ---
interface Ethernet0/1
description Link_to_Emergency_Firewall
no switchport
ip address 10.1.100.1 255.255.255.252
no shutdown
```

```
!—-------------------------------------------------------------
!Rutare:


router ospf 1
 router-id 10.1.100.1
 network 10.1.5.0 0.0.0.255 area 1
 network 10.1.10.0 0.0.0.255 area 1
 network 10.1.20.0 0.0.0.255 area 1
 network 10.1.30.0 0.0.0.255 area 1
 network 10.1.40.0 0.0.0.255 area 1
 network 10.1.100.0 0.0.0.3 area 1
 passive-interface Vlan5
 passive-interface Vlan10
 passive-interface Vlan20
 passive-interface Vlan30
 passive-interface Vlan40
exit

do write

!—-------------------------------------------------------------
!DHCP:

interface range Vlan10 , Vlan20 , Vlan30 , Vlan40
 ip helper-address 192.168.20.10
exit
```

EB-SW1

```
en
conf t
host EB-SW1

no ip routing
spanning-tree mode rapid-pvst

! --- 1. Definire VLAN-uri Emergency Building ---
vlan 5
 name Management
vlan 10
 name ICU
vlan 20
 name Radiology
vlan 30
 name Triage
vlan 40
 name Local-Server
vlan 99
 name Native
exit

! --- 2. Porturi de Access (Conform Topologiei din GNS3) ---
interface Ethernet0/0
 description Access_ICU_VLAN10
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

interface Ethernet0/1
 description Access_Radiology_VLAN20
 switchport mode access
 switchport access vlan 20
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

interface Ethernet0/2
 description Access_Triage_VLAN30
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

! --- 3. Pregătire Interfețe Trunk (Fizice) ---
```

```
! Doar e1/0 și e1/1 pleacă spre Core-Switch
interface range e1/0-1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 channel-group 1 mode active
 no shutdown
 exit

! --- 4. Interfața Logică Port-channel 1 ---
interface Port-channel1
 description Trunk_to_EB-SW-CORE
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 no shutdown
 exit

! --- 5. IP de Management (VLAN 5) ---
interface Vlan5
 description Management_IP
 ip address 10.1.5.11 255.255.255.0
 no shutdown
 exit

ip default-gateway 10.1.5.1
```

!____________________________________

!Port security:

```
configure terminal
ip dhcp snooping
no ip dhcp snooping information option
ip dhcp snooping vlan 10,20,30

interface range Ethernet0/0 - 2
 switchport port-security
 switchport port-security maximum 2
 switchport port-security violation shutdown
 switchport port-security mac-address sticky
 ip dhcp snooping limit rate 15

interface Port-channel1
 ip dhcp snooping trust
exit
write memory
```

EB-SW2

```
en
conf t
host EB-SW2

no ip routing
spanning-tree mode rapid-pvst

! --- 1. Definire VLAN-uri Emergency Building ---
vlan 5
 name Management
vlan 10
 name ICU
vlan 20
 name Radiology
vlan 30
 name Triage
vlan 40
 name Local-Server
vlan 99
 name Native
exit

! --- 2. Porturi de Access (Conform Topologiei din GNS3) ---
interface Ethernet0/0
 description Access_Local-Server_VLAN40
 switchport mode access
 switchport access vlan 40
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

! --- 3. Pregătire Interfețe Trunk (Fizice) ---
! Porturile e1/2 și e1/3 pleacă spre Core-Switch
interface range e1/2-3
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 channel-group 1 mode active
 no shutdown
 exit

! --- 4. Interfața Logică Port-channel 1 ---
interface Port-channel1
 description Trunk_to_EB-SW-CORE
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
```

```
 switchport trunk allowed vlan 5,10,20,30,40,99
 no shutdown
 exit

! --- 5. IP de Management (VLAN 5) ---
interface Vlan5
 description Management_IP
 ip address 10.1.5.12 255.255.255.0
 no shutdown
 exit

ip default-gateway 10.1.5.1
```

!______________________________________

!Port security:

```
configure terminal
ip dhcp snooping
no ip dhcp snooping information option
ip dhcp snooping vlan 40

interface Ethernet0/0
 switchport port-security
 switchport port-security maximum 2
 switchport port-security violation shutdown
 switchport port-security mac-address sticky
 ip dhcp snooping limit rate 15

interface Port-channel1
 ip dhcp snooping trust
exit
write memory
```

Local Server

Medical Lab

ML-R1

```
en
conf t
host ML-R1

int e0/0
 description Link_to_ML-FW1_Outside
 ip add 10.2.102.1 255.255.255.252
 no sh

int e0/2
 description Link_to_ISP
 ip add 203.0.113.14 255.255.255.252
 no sh

exit


!—-------------------------------------------------------
!Rutare:

ip route 0.0.0.0 0.0.0.0 203.0.113.13

router ospf 1
 router-id 12.1.1.1
 network 10.2.102.0 0.0.0.3 area 2
exit


!__________________________________________________
!Nat:


interface Ethernet0/0
 ip nat inside
interface Ethernet0/2
 ip nat outside
exit

access-list 1 permit 10.2.0.0 0.0.255.255
ip nat inside source list 1 interface Ethernet0/2 overload

!__________________________________________________
!Tunel:

crypto isakmp policy 10
 encryption aes 256
 hash sha256
 authentication pre-share
```

```
 group 14
exit

crypto isakmp key SpitalSecurizat123! address 203.0.113.2
crypto isakmp key SpitalSecurizat123! address 203.0.113.6

crypto ipsec transform-set TS-GRE esp-aes esp-sha256-hmac
 mode tunnel
exit

crypto ipsec profile IPSEC-PROF
 set transform-set TS-GRE
exit

! Tunel spre MB-R1 (Calea Primară)
interface Tunnel12
 description Tunnel_to_MB-R1_Primary
 ip address 10.254.12.2 255.255.255.252
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/2
 tunnel destination 203.0.113.2
 tunnel protection ipsec profile IPSEC-PROF
 no shutdown

! Tunel spre MB-R2 (Calea Secundară)
interface Tunnel22
 description Tunnel_to_MB-R2_Backup
 ip address 10.254.22.2 255.255.255.252
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/2
 tunnel destination 203.0.113.6
 tunnel protection ipsec profile IPSEC-PROF
 no shutdown
exit

router ospf 1
 network 10.254.12.0 0.0.0.3 area 0
 network 10.254.22.0 0.0.0.3 area 0
exit
```

ML-FW1

```
enable
cisco
configure terminal
hostname ML-FW1

! --- 1. Configurare Interfață Interioară (Spre ML-SW-CORE) ---
interface GigabitEthernet0/1
 description Link_to_ML-SW-CORE
 nameif inside
 security-level 100
 ip address 10.2.100.2 255.255.255.252
 no shutdown
 exit

! --- 2. Configurare Interfață Exterioară (Spre ML-R1) ---
interface GigabitEthernet0/0
 description Link_to_ML-R1
 nameif outside
 security-level 0
 ip address 10.2.102.2 255.255.255.252
 no shutdown
 exit

! --- 3. Activare Inspecție ICMP (Modular Policy Framework) ---
policy-map global_policy
 class inspection_default
  inspect icmp
 exit
exit

! --- 4. Rutare Implicită (Spre Internet via ML-R1) ---
route outside 0.0.0.0 0.0.0.0 10.2.102.1 1

! --- 5. Motorul de Rutare Dinamică OSPF Area 2 ---
router ospf 1
 router-id 10.2.102.2
 network 10.2.100.0 255.255.255.252 area 2
 network 10.2.102.0 255.255.255.252 area 2
 default-information originate always
exit
```

```
!________________________________________
!ACL:

access-list OUTSIDE_IN extended permit ospf host 10.2.102.1 any
access-list OUTSIDE_IN extended permit icmp 192.168.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.1.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.2.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.3.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.4.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit tcp 192.168.0.0 255.255.0.0 any eq 22
access-group OUTSIDE_IN in interface outside

ssh 0.0.0.0 0.0.0.0 outside
```

ML-SW-CORE

```
en
conf t
host ML-SW-CORE

! --- 1. Activare Rutare L3 și Rapid Spanning Tree ---
ip routing
spanning-tree mode rapid-pvst
spanning-tree vlan 5,10,20,30,40 priority 4096

! --- 2. Definire Bază de Date VLAN-uri ---
vlan 5
 name Management
vlan 10
 name LabPC
vlan 20
 name PCRMachine
vlan 30
 name Triage
vlan 40
 name Local-Database
vlan 99
 name Native
exit

! --- 3. Legătură Downstream către EB-SW1 (Port-channel 1) ---
interface range e1/0-1
 description Link_to_ML-SW1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 channel-group 1 mode active
 no shutdown
 exit

interface Port-channel1
 description Trunk_to_ML-SW1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 no shutdown
 exit

! --- 4. Legătură Downstream către EB-SW2 (Port-channel 2) ---
interface range e1/2-3
 description Link_to_ML-SW2
 switchport trunk encapsulation dot1q
```

```
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 channel-group 2 mode active
 no shutdown
 exit

interface Port-channel2
 description Trunk_to_ML-SW2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 no shutdown
 exit

! --- 5. Configurare SVI-uri (Default Gateways pentru VLAN-uri) ---
interface Vlan5
 description Gateway_Management
 ip address 10.2.5.1 255.255.255.0
 no shutdown

interface Vlan10
 description Gateway_LabPC
 ip address 10.2.10.1 255.255.255.0
 no shutdown

interface Vlan20
 description Gateway_PCRMachine
 ip address 10.2.20.1 255.255.255.0
 no shutdown

interface Vlan30
 description Gateway_Analyzer
 ip address 10.2.30.1 255.255.255.0
 no shutdown

interface Vlan40
 description Gateway_Local-Database
 ip address 10.2.40.1 255.255.255.0
 no shutdown

! --- 6. Pregătire Port către Firewall ---
interface Ethernet0/1
description Link_to_Medical_Lab_Firewall
no switchport
ip address 10.2.100.1 255.255.255.252
no shutdown
```

```
!—-------------------------------------------------------
!Rutare:


router ospf 1
 router-id 10.2.100.1
 network 10.2.5.0 0.0.0.255 area 2
 network 10.2.10.0 0.0.0.255 area 2
 network 10.2.20.0 0.0.0.255 area 2
 network 10.2.30.0 0.0.0.255 area 2
 network 10.2.40.0 0.0.0.255 area 2
 network 10.2.100.0 0.0.0.3 area 2
 passive-interface Vlan5
 passive-interface Vlan10
 passive-interface Vlan20
 passive-interface Vlan30
 passive-interface Vlan40
exit


!—-------------------------------------------------------
!DHCP::

interface range Vlan10 , Vlan20 , Vlan30 , Vlan40
 ip helper-address 192.168.20.10
exit
```

ML-SW1

```
en
conf t
host ML-SW1

no ip routing
spanning-tree mode rapid-pvst

! --- 1. Definire VLAN-uri Emergency Building ---
vlan 5
 name Management
vlan 10
 name LabPC
vlan 20
 name PCRMachine
vlan 30
 name Triage
vlan 40
 name Local-Database
vlan 99
 name Native
exit

! --- 2. Porturi de Access (Conform Topologiei din GNS3) ---
interface Ethernet0/0
 description Access_LabPC_VLAN10
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

interface Ethernet0/1
 description Access_PCR_VLAN20
 switchport mode access
 switchport access vlan 20
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

interface Ethernet0/2
 description Access_Analyzer_VLAN30
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

! --- 3. Pregătire Interfețe Trunk (Fizice) ---
```

```
! Doar e1/0 și e1/1 pleacă spre Core-Switch
interface range e1/0-1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 channel-group 1 mode active
 no shutdown
 exit

! --- 4. Interfața Logică Port-channel 1 ---
interface Port-channel1
 description Trunk_to_ML-SW-CORE
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 no shutdown
 exit

! --- 5. IP de Management (VLAN 5) ---
interface Vlan5
 description Management_IP
 ip address 10.2.5.11 255.255.255.0
 no shutdown
 exit

ip default-gateway 10.2.5.1
```

!____

!Port security:

```
configure terminal
ip dhcp snooping
no ip dhcp snooping information option
ip dhcp snooping vlan 10,20,30

interface range Ethernet0/0 - 2
 switchport port-security
 switchport port-security maximum 2
 switchport port-security violation shutdown
 switchport port-security mac-address sticky
 ip dhcp snooping limit rate 15

interface Port-channel1
 ip dhcp snooping trust
exit
```

write memory

ML-SW2

```
en
conf t
host ML-SW2

no ip routing
spanning-tree mode rapid-pvst

! --- 1. Definire VLAN-uri Emergency Building ---
vlan 5
 name Management
vlan 10
 name LabPC
vlan 20
 name PCRMachine
vlan 30
 name Triage
vlan 40
 name Local-Database
vlan 99
 name Native
exit

! --- 2. Porturi de Access (Conform Topologiei din GNS3) ---
interface Ethernet0/0
 description Access_Local-Databaser_VLAN40
 switchport mode access
 switchport access vlan 40
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

! --- 3. Pregătire Interfețe Trunk (Fizice) ---
! Porturile e1/2 și e1/3 pleacă spre Core-Switch
interface range e1/2-3
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 channel-group 1 mode active
 no shutdown
 exit

! --- 4. Interfața Logică Port-channel 1 ---
interface Port-channel1
 description Trunk_to_ML-SW-CORE
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
```

```
 switchport trunk allowed vlan 5,10,20,30,40,99
 no shutdown
 exit

! --- 5. IP de Management (VLAN 5) ---
interface Vlan5
 description Management_IP
 ip address 10.2.5.12 255.255.255.0
 no shutdown
 exit

ip default-gateway 10.2.5.1
```

```
configure terminal
ip dhcp snooping
no ip dhcp snooping information option
ip dhcp snooping vlan 40

interface Ethernet0/0
 switchport port-security
 switchport port-security maximum 2
 switchport port-security violation shutdown
 switchport port-security mac-address sticky
 ip dhcp snooping limit rate 15

interface Port-channel1
 ip dhcp snooping trust
exit
write memory
```


```
configure terminal
ip dhcp snooping
no ip dhcp snooping information option
ip dhcp snooping vlan 40

interface Ethernet0/0
 switchport port-security
 switchport port-security maximum 2
 switchport port-security violation shutdown
 switchport port-security mac-address sticky
 ip dhcp snooping limit rate 15

interface Port-channel1
 ip dhcp snooping trust
exit
write memory
```

Local Server1

Clinic

CL-R2

```
en
conf t
host CL-R2

int e0/0
 description Link_to_CL-FW1_Outside
 ip add 10.3.102.1 255.255.255.252
 no sh

int e0/3
 description Link_to_ISP
 ip add 203.0.113.18 255.255.255.252
 no sh

exit


!—------------------------------------------------------
!Rutare:

ip route 0.0.0.0 0.0.0.0 203.0.113.17

router ospf 1
 router-id 13.1.1.1
 network 10.3.102.0 0.0.0.3 area 3
exit


!___________________________________________
!Nat:


interface Ethernet0/0
 ip nat inside
interface Ethernet0/3
 ip nat outside
exit

access-list 1 permit 10.3.0.0 0.0.255.255
ip nat inside source list 1 interface Ethernet0/3 overload


!___________________________________________
!Tunel:


crypto isakmp policy 10
 encryption aes 256
```

```
 hash sha256
 authentication pre-share
 group 14
exit

crypto isakmp key SpitalSecurizat123! address 203.0.113.2
crypto isakmp key SpitalSecurizat123! address 203.0.113.6

crypto ipsec transform-set TS-GRE esp-aes esp-sha256-hmac
 mode tunnel
exit

crypto ipsec profile IPSEC-PROF
 set transform-set TS-GRE
exit

! Tunel spre MB-R1 (Calea Primară)
interface Tunnel13
 description Tunnel_to_MB-R1_Primary
 ip address 10.254.13.2 255.255.255.252
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/3
 tunnel destination 203.0.113.2
 tunnel protection ipsec profile IPSEC-PROF
 no shutdown

! Tunel spre MB-R2 (Calea Secundară)
interface Tunnel23
 description Tunnel_to_MB-R2_Backup
 ip address 10.254.23.2 255.255.255.252
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/3
 tunnel destination 203.0.113.6
 tunnel protection ipsec profile IPSEC-PROF
 no shutdown
exit

router ospf 1
 network 10.254.13.0 0.0.0.3 area 0
 network 10.254.23.0 0.0.0.3 area 0
exit
```

CL-FW1

```
enable
cisco
configure terminal
hostname CL-FW1

! --- 1. Configurare Interfață Interioară (Spre CL-SW-CORE) ---
interface GigabitEthernet0/1
 description Link_to_CL-SW-CORE
 nameif inside
 security-level 100
 ip address 10.3.100.2 255.255.255.252
 no shutdown
 exit

! --- 2. Configurare Interfață Exterioară (Spre CL-R2) ---
interface GigabitEthernet0/0
 description Link_to_CL-FW1_Outside
 nameif outside
 security-level 0
 ip address 10.3.102.2 255.255.255.252
 no shutdown
 exit

! --- 3. Activare Inspecție ICMP (Modular Policy Framework) ---
policy-map global_policy
 class inspection_default
  inspect icmp
 exit
exit

! --- 4. Rutare Implicită (Spre Internet via CL-R2) ---
route outside 0.0.0.0 0.0.0.0 10.3.102.1 1

! --- 5. Motorul de Rutare Dinamică OSPF Area 3 ---
router ospf 1
 router-id 10.3.102.2
 network 10.3.100.0 255.255.255.252 area 3
 network 10.3.102.0 255.255.255.252 area 3
 default-information originate always
exit
```

```
!_______________________________________
!ACL:

access-list OUTSIDE_IN extended permit ospf host 10.3.102.1 any
access-list OUTSIDE_IN extended permit icmp 192.168.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.1.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.2.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.3.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.4.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit tcp 192.168.0.0 255.255.0.0 any eq 22
access-group OUTSIDE_IN in interface outside


ssh 0.0.0.0 0.0.0.0 outside
```

CL-SW-CORE

```
en
conf t
host CL-SW-CORE

! --- 1. Activare Rutare L3 și Rapid Spanning Tree ---
ip routing
spanning-tree mode rapid-pvst
spanning-tree vlan 5,10,20,30,40 priority 4096

! --- 2. Definire Bază de Date VLAN-uri ---
vlan 5
 name Management
vlan 10
 name DoctorPC
vlan 20
 name ReceptionPC
vlan 30
 name NurseTerminal
vlan 40
 name Local-Server1
vlan 99
 name Native
exit

! --- 3. Legătură Downstream către EB-SW1 (Port-channel 1) ---
interface range e1/0-1
 description Link_to_CL-SW1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 channel-group 1 mode active
 no shutdown
 exit

interface Port-channel1
 description Trunk_to_CL-SW1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 no shutdown
 exit

! --- 4. Legătură Downstream către EB-SW2 (Port-channel 2) ---
interface range e1/2-3
 description Link_to_CL-SW2
 switchport trunk encapsulation dot1q
```

```
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 channel-group 2 mode active
 no shutdown
 exit

interface Port-channel2
 description Trunk_to_CL-SW2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 no shutdown
 exit

! --- 5. Configurare SVI-uri (Default Gateways pentru VLAN-uri) ---
interface Vlan5
 description Gateway_Management
 ip address 10.3.5.1 255.255.255.0
 no shutdown

interface Vlan10
 description Gateway_DoctorPC
 ip address 10.3.10.1 255.255.255.0
 no shutdown

interface Vlan20
 description Gateway_ReceptionPC
 ip address 10.3.20.1 255.255.255.0
 no shutdown

interface Vlan30
 description Gateway_NurseTerminal
 ip address 10.3.30.1 255.255.255.0
 no shutdown

interface Vlan40
 description Gateway_Local-Server1
 ip address 10.3.40.1 255.255.255.0
 no shutdown

! --- 6. Pregătire Port către Firewall ---
interface Ethernet0/1
description Link_to_Clinic_Lab_Firewall
no switchport
ip address 10.3.100.1 255.255.255.252
no shutdown
```

```
!—------------------------------------------------------
!Rutare:



router ospf 1
 router-id 10.3.100.1
 network 10.3.5.0 0.0.0.255 area 3
 network 10.3.10.0 0.0.0.255 area 3
 network 10.3.20.0 0.0.0.255 area 3
 network 10.3.30.0 0.0.0.255 area 3
 network 10.3.40.0 0.0.0.255 area 3
 network 10.3.100.0 0.0.0.3 area 3
 passive-interface Vlan5
 passive-interface Vlan10
 passive-interface Vlan20
 passive-interface Vlan30
 passive-interface Vlan40
exit


!—------------------------------------------------------
!DHCP::

interface range Vlan10 , Vlan20 , Vlan30 , Vlan40
 ip helper-address 192.168.20.10
exit
```

CL-SW1

```
en
conf t
host CL-SW1

no ip routing
spanning-tree mode rapid-pvst

! --- 1. Definire VLAN-uri Emergency Building ---
vlan 5
 name Management
vlan 10
 name DoctorPC
vlan 20
 name ReceptionPC
vlan 30
 name NurseTerminal
vlan 40
 name Local-Server1
vlan 99
 name Native
exit

! --- 2. Porturi de Access (Conform Topologiei din GNS3) ---
interface Ethernet0/0
 description Access_DoctorPC_VLAN10
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

interface Ethernet0/1
 description Access_ReceptionPC_VLAN20
 switchport mode access
 switchport access vlan 20
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

interface Ethernet0/2
 description Access_NurseTerminal_VLAN30
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

! --- 3. Pregătire Interfețe Trunk (Fizice) ---
```

```
! Doar e1/0 și e1/1 pleacă spre Core-Switch
interface range e1/0-1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 channel-group 1 mode active
 no shutdown
 exit

! --- 4. Interfața Logică Port-channel 1 ---
interface Port-channel1
 description Trunk_to_CL-SW-CORE
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 no shutdown
 exit

! --- 5. IP de Management (VLAN 5) ---
interface Vlan5
 description Management_IP
 ip address 10.3.5.11 255.255.255.0
 no shutdown
 exit

ip default-gateway 10.3.5.1
```

!____

!Port security:

```
configure terminal
ip dhcp snooping
no ip dhcp snooping information option
ip dhcp snooping vlan 10,20,30

interface range Ethernet0/0 - 2
 switchport port-security
 switchport port-security maximum 2
 switchport port-security violation shutdown
 switchport port-security mac-address sticky
 ip dhcp snooping limit rate 15

interface Port-channel1
 ip dhcp snooping trust
exit
Do write memory
```

CL-SW2

```
en
conf t
host CL-SW2

no ip routing
spanning-tree mode rapid-pvst

! --- 1. Definire VLAN-uri Emergency Building ---
vlan 5
 name Management
vlan 10
 name DoctorPC
vlan 20
 name ReceptionPC
vlan 30
 name NurseTerminal
vlan 40
 name Local-Server1
vlan 99
 name Native
exit

! --- 2. Porturi de Access (Conform Topologiei din GNS3) ---
interface Ethernet0/0
 description Access_Local-Server1_VLAN40
 switchport mode access
 switchport access vlan 40
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

! --- 3. Pregătire Interfețe Trunk (Fizice) ---
! Porturile e1/2 și e1/3 pleacă spre Core-Switch
interface range e1/2-3
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 channel-group 1 mode active
 no shutdown
 exit

! --- 4. Interfața Logică Port-channel 1 ---
interface Port-channel1
 description Trunk_to_CL-SW-CORE
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
```

```
 switchport trunk allowed vlan 5,10,20,30,40,99
 no shutdown
 exit

! --- 5. IP de Management (VLAN 5) ---
interface Vlan5
 description Management_IP
 ip address 10.3.5.12 255.255.255.0
 no shutdown
 exit

ip default-gateway 10.3.5.1
```

!_______________________________________
!Port security:

```
configure terminal
ip dhcp snooping
no ip dhcp snooping information option
ip dhcp snooping vlan 40

interface Ethernet0/0
 switchport port-security
 switchport port-security maximum 2
 switchport port-security violation shutdown
 switchport port-security mac-address sticky
 ip dhcp snooping limit rate 15

interface Port-channel1
 ip dhcp snooping trust
exit
write memory
```

Local Server2

Remote Site

RS-R1

```
en
conf t
host RS-R1

int e0/0
 description Link_to_RS-FW1_Outside
 ip add 10.4.102.1 255.255.255.252
 no sh

int e0/1
 description Link_to_RS-R2_CrossConnect
 ip add 10.4.104.1 255.255.255.252
 no sh

int e0/2
 description Link_to_ISP
 ip add 203.0.113.22 255.255.255.252
 no sh

exit


!—-----------------------------------------------------------
!Rutare:

ip route 0.0.0.0 0.0.0.0 203.0.113.21

router ospf 1
router-id 14.1.1.1
network 10.4.102.0 0.0.0.3 area 4
network 10.4.104.0 0.0.0.3 area 4
exit


!____________________________________________
!Nat:


interface Ethernet0/0
 ip nat inside
interface Ethernet0/1
 ip nat inside
interface Ethernet0/2
 ip nat outside
exit

access-list 1 permit 10.4.0.0 0.0.255.255
ip nat inside source list 1 interface Ethernet0/2 overload
```

```
!___________________________________________
!Tunel:

crypto isakmp policy 10
 encryption aes 256
 hash sha256
 authentication pre-share
 group 14
exit

crypto isakmp key SpitalSecurizat123! address 203.0.113.2
crypto isakmp key SpitalSecurizat123! address 203.0.113.6

crypto ipsec transform-set TS-GRE esp-aes esp-sha256-hmac
 mode tunnel
exit

crypto ipsec profile IPSEC-PROF
 set transform-set TS-GRE
exit

! Tunel spre MB-R1 (Calea Primară via RS-R1)
interface Tunnel14
 description Tunnel_to_MB-R1_Primary
 ip address 10.254.14.2 255.255.255.252
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/2
 tunnel destination 203.0.113.2
 tunnel protection ipsec profile IPSEC-PROF
 no shutdown

! Tunel spre MB-R2 (Calea Secundară via RS-R1)
interface Tunnel24
 description Tunnel_to_MB-R2_Backup
 ip address 10.254.24.2 255.255.255.252
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/2
 tunnel destination 203.0.113.6
 tunnel protection ipsec profile IPSEC-PROF
 no shutdown
exit

router ospf 1
 network 10.254.14.0 0.0.0.3 area 0
```

```
 network 10.254.24.0 0.0.0.3 area 0
exit
```

RS-R2

```
en
conf t
host RS-R2

int e0/0
 description Link_to_RS-FW2_Outside
 ip add 10.4.103.1 255.255.255.252
 no sh

int e0/1
 description Link_to_RS-R1_CrossConnect
 ip add 10.4.104.2 255.255.255.252
 no sh

int e0/3
 description Link_to_ISP
 ip add 203.0.113.26 255.255.255.252
 no sh

exit


!—--------------------------------------------------------
!Rutare:

ip route 0.0.0.0 0.0.0.0 203.0.113.25

router ospf 1
router-id 14.2.2.2
network 10.4.103.0 0.0.0.3 area 4
 network 10.4.104.0 0.0.0.3 area 4
 exit


!___________________________________________
!Nat:


interface Ethernet0/0
 ip nat inside
interface Ethernet0/1
 ip nat inside
interface Ethernet0/3
 ip nat outside
exit

access-list 1 permit 10.4.0.0 0.0.255.255
ip nat inside source list 1 interface Ethernet0/3 overload
```

```
!_______________________________________________
!Tunel:

crypto isakmp policy 10
 encryption aes 256
 hash sha256
 authentication pre-share
 group 14
exit

crypto isakmp key SpitalSecurizat123! address 203.0.113.2
crypto isakmp key SpitalSecurizat123! address 203.0.113.6

crypto ipsec transform-set TS-GRE esp-aes esp-sha256-hmac
 mode tunnel
exit

crypto ipsec profile IPSEC-PROF
 set transform-set TS-GRE
exit

! Tunel spre MB-R1 (Calea Primară via RS-R2)
interface Tunnel15
 description Tunnel_to_MB-R1_Primary
 ip address 10.254.15.2 255.255.255.252
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/3
 tunnel destination 203.0.113.2
 tunnel protection ipsec profile IPSEC-PROF
 no shutdown

! Tunel spre MB-R2 (Calea Secundară via RS-R2)
interface Tunnel25
 description Tunnel_to_MB-R2_Backup
 ip address 10.254.25.2 255.255.255.252
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/3
 tunnel destination 203.0.113.6
 tunnel protection ipsec profile IPSEC-PROF
 no shutdown
exit

router ospf 1
 network 10.254.15.0 0.0.0.3 area 0
```

```
 network 10.254.25.0 0.0.0.3 area 0
exit
```

RS-FW1

```
enable
cisco
configure terminal
hostname RS-FW1

! Curățăm interfețele conform scriptului de bază
interface GigabitEthernet0/1
 nameif inside
 security-level 100
 ip address 10.4.100.2 255.255.255.252
 no shutdown
exit

interface GigabitEthernet0/0
 nameif outside
 security-level 0
 ip address 10.4.102.2 255.255.255.252
 no shutdown
exit

policy-map global_policy
 class inspection_default
 no  inspect icmp
Exit
exit

!—-------------------------------------------------------
!Rutare:

route outside 0.0.0.0 0.0.0.0 10.4.102.1 1

router ospf 1
 router-id 10.4.102.2
 network 10.4.100.0 255.255.255.252 area 4
 network 10.4.102.0 255.255.255.252 area 4
 default-information originate always
exit

!________________________________________
!ACL:
access-list OUTSIDE_IN extended permit ospf host 10.4.102.1 any
access-list OUTSIDE_IN extended permit icmp 192.168.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.1.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.2.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.3.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.4.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp any 10.4.0.0 255.255.0.0
```

```
access-group OUTSIDE_IN in interface outside


!_________________________
!Ansible:


access-list OUTSIDE_IN extended permit tcp 192.168.0.0 255.255.0.0 any eq 22

access-list TCP_BYPASS extended permit tcp 192.168.0.0 255.255.0.0 any eq 22

class-map TCP_CLASS
 match access-list TCP_BYPASS
exit

policy-map global_policy
 class TCP_CLASS
  set connection advanced-options tcp-state-bypass
Exit
exit
write memory


access-list TCP_BYPASS extended permit tcp any eq 22 192.168.0.0 255.255.0.0

ssh 0.0.0.0 0.0.0.0 outside
```

RS-FW2

```
enable
cisco
configure terminal
hostname RS-FW2

interface GigabitEthernet0/1
 nameif inside
 security-level 100
 ip address 10.4.100.6 255.255.255.252
 no shutdown
Exit

interface GigabitEthernet0/0
 nameif outside
 security-level 0
 ip address 10.4.103.2 255.255.255.252
 no shutdown
exit

policy-map global_policy
 class inspection_default
no  inspect icmp
Exit
exit

!—-------------------------------------------------------
!Rutare:

route outside 0.0.0.0 0.0.0.0 10.4.103.1 1

router ospf 1
 router-id 10.4.103.2
 network 10.4.100.4 255.255.255.252 area 4
 network 10.4.103.0 255.255.255.252 area 4
 default-information originate always
exit

!__________________________________________
!ACL:

access-list OUTSIDE_IN extended permit ospf host 10.4.103.1 any
access-list OUTSIDE_IN extended permit icmp 192.168.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.1.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.2.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.3.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp 10.4.0.0 255.255.0.0 any
access-list OUTSIDE_IN extended permit icmp any 10.4.0.0 255.255.0.0
access-group OUTSIDE_IN in interface outside
```

```
access-list OUTSIDE_IN extended permit tcp 192.168.0.0 255.255.0.0 any eq 22

access-list TCP_BYPASS extended permit tcp 192.168.0.0 255.255.0.0 any eq 22

class-map TCP_CLASS
 match access-list TCP_BYPASS
exit

policy-map global_policy
 class TCP_CLASS
  set connection advanced-options tcp-state-bypass
Exit
exit
write memory

access-list TCP_BYPASS extended permit tcp any eq 22 192.168.0.0 255.255.0.0

ssh 0.0.0.0 0.0.0.0 outside
```

RS-SW1

```
en
conf t
host RS-SW1

ip routing
spanning-tree mode rapid-pvst
spanning-tree vlan 5,10,20,30 priority 4096
spanning-tree vlan 40,50,60 priority 8192

! --- 2. Definire VLAN-uri ---
vlan 5
 name Management
vlan 10
 name Doctor
vlan 20
 name MRI
vlan 30
 name X-RAY
vlan 40
 name Server
vlan 50
 name HRPC
vlan 60
 name Admin
vlan 99
 name Native
exit

! --- 3. Pregătire Interfețe Trunk (Fizice) ---
! Porturile e1/0 și e1/1 pleacă spre Acces-Switch-1 (RS-SW3)
interface range e1/0-1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,50,60,99
 channel-group 1 mode active
 no shutdown
 exit

! --- 4. Interfața Logică Port-channel 1 ---
interface Port-channel1
 description Trunk_to_RS-SW3
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,50,60,99
 no shutdown
 exit
```

```
! Porturile e1/2 și e1/3 pleacă spre Acces-Switch-2 (RS-SW4)
interface range e1/2-3
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,50,60,99
 channel-group 2 mode active
 no shutdown
 exit

! --- 5. Interfața Logică Port-channel 2 ---
interface Port-channel2
 description Trunk_to_RS-SW4
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,50,60,99
 no shutdown
 exit

! --- 6. Configurare SVI-uri cu HSRP (V2) ---
! RS-SW1 este Active (priority 150 + preempt) pentru 5, 10, 20, 30
interface Vlan5
 description Gateway_Management
 ip address 10.4.5.2 255.255.255.0
 standby version 2
 standby 5 ip 10.4.5.1
 standby 5 priority 150
 standby 5 preempt
 no shutdown

interface Vlan10
 description Gateway_Doctor
 ip address 10.4.10.2 255.255.255.0
 standby version 2
 standby 10 ip 10.4.10.1
 standby 10 priority 150
 standby 10 preempt
 no shutdown

interface Vlan20
 description Gateway_MRI
 ip address 10.4.20.2 255.255.255.0
 standby version 2
 standby 20 ip 10.4.20.1
 standby 20 priority 150
 standby 20 preempt
```

```
 no shutdown

interface Vlan30
 description Gateway_X-RAY
 ip address 10.4.30.2 255.255.255.0
 standby version 2
 standby 30 ip 10.4.30.1
 standby 30 priority 150
 standby 30 preempt
 no shutdown

! RS-SW1 este Standby (priority 100 implicit) pentru 40, 50, 60
interface Vlan40
 description Gateway_Server
 ip address 10.4.40.2 255.255.255.0
 standby version 2
 standby 40 ip 10.4.40.1
 standby 40 priority 100
 no shutdown

interface Vlan50
 description Gateway_HRPC
 ip address 10.4.50.2 255.255.255.0
 standby version 2
 standby 50 ip 10.4.50.1
 standby 50 priority 100
 no shutdown

interface Vlan60
 description Gateway_Admin
 ip address 10.4.60.2 255.255.255.0
 standby version 2
 standby 60 ip 10.4.60.1
 standby 60 priority 100
 no shutdown


!Extern spre Firewall
interface Ethernet0/1
description Link_to_RS-FW1
no switchport
ip address 10.4.100.1 255.255.255.252
no shutdown
exit
```

```
!—--------------------------------------------------------
!Rutare:


router ospf 1
 router-id 10.4.100.1
 network 10.4.5.0 0.0.0.255 area 4
 network 10.4.10.0 0.0.0.255 area 4
 network 10.4.20.0 0.0.0.255 area 4
 network 10.4.30.0 0.0.0.255 area 4
 network 10.4.40.0 0.0.0.255 area 4
 network 10.4.50.0 0.0.0.255 area 4
 network 10.4.60.0 0.0.0.255 area 4
 network 10.4.100.0 0.0.0.3 area 4
 passive-interface Vlan5
 passive-interface Vlan10
 passive-interface Vlan20
 passive-interface Vlan30
 passive-interface Vlan40
 passive-interface Vlan50
 passive-interface Vlan60
exit
```

RS-SW2

```
en
conf t
host RS-SW2

ip routing
spanning-tree mode rapid-pvst
spanning-tree vlan 40,50,60 priority 4096
spanning-tree vlan 5,10,20,30 priority 8192

! --- 2. Definire VLAN-uri ---
vlan 5
 name Management
vlan 10
 name Doctor
vlan 20
 name MRI
vlan 30
 name X-RAY
vlan 40
 name Server
vlan 50
 name HRPC
vlan 60
 name Admin
vlan 99
 name Native
exit


! --- 4. Pregătire Interfețe Trunk (Fizice) ---
interface range e1/0-1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,50,60,99
 channel-group 1 mode active
 no shutdown
 exit

! --- 5. Interfața Logică Port-channel 1 ---
interface Port-channel1
 description Trunk_to_RS-SW4
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,50,60,99
 no shutdown
 exit
```

```
! Porturile e1/2 și e1/3 pleacă spre Access-Switch-2 (RS-SW4)
interface range e1/2-3
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,50,60,99
 channel-group 2 mode active
 no shutdown
 exit

! --- 6. Interfața Logică Port-channel 2 ---
interface Port-channel2
 description Trunk_to_RS-SW3
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,50,60,99
 no shutdown
 exit

! --- 7. Configurare SVI-uri cu HSRP (V2) ---
interface Vlan5
 description Gateway_Management
 ip address 10.4.5.3 255.255.255.0
 standby version 2
 standby 5 ip 10.4.5.1
 standby 5 priority 100
 no shutdown

interface Vlan10
 description Gateway_Doctor
 ip address 10.4.10.3 255.255.255.0
 standby version 2
 standby 10 ip 10.4.10.1
 standby 10 priority 100
 no shutdown

interface Vlan20
 description Gateway_MRI
 ip address 10.4.20.3 255.255.255.0
 standby version 2
 standby 20 ip 10.4.20.1
 standby 20 priority 100
 no shutdown

interface Vlan30
 description Gateway_X-RAY
```

```
 ip address 10.4.30.3 255.255.255.0
 standby version 2
 standby 30 ip 10.4.30.1
 standby 30 priority 100
 no shutdown

! RS-SW2 este Active (priority 150 + preempt) pentru 40, 50, 60
interface Vlan40
 description Gateway_Server
 ip address 10.4.40.3 255.255.255.0
 standby version 2
 standby 40 ip 10.4.40.1
 standby 40 priority 150
 standby 40 preempt
 no shutdown

interface Vlan50
 description Gateway_HRPC
 ip address 10.4.50.3 255.255.255.0
 standby version 2
 standby 50 ip 10.4.50.1
 standby 50 priority 150
 standby 50 preempt
 no shutdown

interface Vlan60
 description Gateway_Admin
 ip address 10.4.60.3 255.255.255.0
 standby version 2
 standby 60 ip 10.4.60.1
 standby 60 priority 150
 standby 60 preempt
 no shutdown


!Interfață Ruterizată spre Firewall (RS-FW2)
interface Ethernet0/1
 description Link_to_RS-FW2
 no switchport
 ip address 10.4.100.5 255.255.255.252
 no shutdown
 Exit
```

```
!—-------------------------------------------------------
!Rutare:

router ospf 1
 router-id 10.4.100.5
 network 10.4.5.0 0.0.0.255 area 4
 network 10.4.10.0 0.0.0.255 area 4
 network 10.4.20.0 0.0.0.255 area 4
 network 10.4.30.0 0.0.0.255 area 4
 network 10.4.40.0 0.0.0.255 area 4
 network 10.4.50.0 0.0.0.255 area 4
 network 10.4.60.0 0.0.0.255 area 4
 network 10.4.100.4 0.0.0.3 area 4
 passive-interface Vlan5
 passive-interface Vlan10
 passive-interface Vlan20
 passive-interface Vlan30
 passive-interface Vlan40
 passive-interface Vlan50
 passive-interface Vlan60
exit
```

RS-SW3

```
en
conf t
host RS-SW3

no ip routing
spanning-tree mode rapid-pvst

! --- 1. Definire VLAN-uri Emergency Building ---
vlan 5
 name Management
vlan 10
 name Doctor
vlan 20
 name MRI
vlan 30
 name X-RAY
vlan 40
 name Server
vlan 50
 name HRPC
vlan 60
 name Admin
vlan 99
 name Native
exit

! --- 2. Porturi de Access (Conform Topologiei din GNS3) ---
interface Ethernet0/0
 description Access_Doctor_VLAN10
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

interface Ethernet0/1
 description Access_MRI_VLAN20
 switchport mode access
 switchport access vlan 20
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

interface Ethernet0/2
 description Access_X-RAY_VLAN30
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast
```

```
 spanning-tree bpduguard enable
 no shutdown

! --- 3. Pregătire Interfețe Trunk (Fizice) ---
! Porturile e1/0 și e1/1 pleacă spre Core-Switch-1 (RS-SW1)
interface range e1/0-1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,50,60,99
 channel-group 1 mode active
 no shutdown
 exit

! --- 4. Interfața Logică Port-channel 1 ---
interface Port-channel1
 description Trunk_to_RS-SW1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,50,60,99
 no shutdown
 exit

! Porturile e1/2 și e1/3 pleacă spre Core-Switch-2 (RS-SW2)
interface range e1/2-3
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,50,60,99
 channel-group 2 mode active
 no shutdown
 exit

! --- 5. Interfața Logică Port-channel 2 ---
interface Port-channel2
 description Trunk_to_RS-SW2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,50,60,99
 no shutdown
 exit

! --- 6. IP de Management (VLAN 5) ---
interface Vlan5
 description Management_IP
 ip address 10.4.5.13 255.255.255.0
```

```
 no shutdown
 exit

ip default-gateway 10.4.5.1
```

!____________________________________

!Port security:

```
configure terminal
ip dhcp snooping
no ip dhcp snooping information option
ip dhcp snooping vlan 10,20,30

interface range Ethernet0/0 - 2
 switchport port-security
 switchport port-security maximum 2
 switchport port-security violation shutdown
 switchport port-security mac-address sticky
 ip dhcp snooping limit rate 15

interface Port-channel1
 ip dhcp snooping trust
interface Port-channel2
 ip dhcp snooping trust
exit
write memory
```

RS-SW4

```
en
conf t
host RS-SW4

no ip routing
spanning-tree mode rapid-pvst

! --- 1. Definire VLAN-uri Emergency Building ---
vlan 5
 name Management
vlan 10
 name Doctor
vlan 20
 name MRI
vlan 30
 name X-RAY
vlan 40
 name Server
vlan 50
 name HRPC
vlan 60
 name Admin
vlan 99
 name Native
exit

! --- 2. Porturi de Access (Conform Topologiei din GNS3) ---
interface Ethernet0/0
 description Access_Server_VLAN60
 switchport mode access
 switchport access vlan 60
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

interface Ethernet0/1
 description Access_HRPC_VLAN50
 switchport mode access
 switchport access vlan 50
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

interface Ethernet0/2
 description Access_Admin_VLAN40
 switchport mode access
 switchport access vlan 40
 spanning-tree portfast
```

```
 spanning-tree bpduguard enable
 no shutdown

! --- 3. Pregătire Interfețe Trunk (Fizice) ---
! Porturile e1/0 și e1/1 pleacă spre Core-Switch-1 (RS-SW1)
interface range e1/0-1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,50,60,99
 channel-group 1 mode active
 no shutdown
 exit

! --- 4. Interfața Logică Port-channel 1 ---
interface Port-channel1
 description Trunk_to_RS-SW1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,50,60,99
 no shutdown
 exit

! Porturile e1/2 și e1/3 pleacă spre Core-Switch-2 (RS-SW2)
interface range e1/2-3
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,50,60,99
 channel-group 2 mode active
 no shutdown
 exit

! --- 5. Interfața Logică Port-channel 2 ---
interface Port-channel2
 description Trunk_to_RS-SW2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,50,60,99
 no shutdown
 exit

! --- 6. IP de Management (VLAN 5) ---
interface Vlan5
 description Management_IP
 ip address 10.4.5.14 255.255.255.0
```

```
 no shutdown
 exit

ip default-gateway 10.4.5.1
```

!____________________________________
!Port security:

```
configure terminal
ip dhcp snooping
no ip dhcp snooping information option
ip dhcp snooping vlan 40,50,60

interface range Ethernet0/0 - 2
 switchport port-security
 switchport port-security maximum 2
 switchport port-security violation shutdown
 switchport port-security mac-address sticky
 ip dhcp snooping limit rate 15

interface Port-channel1
 ip dhcp snooping trust
interface Port-channel2
 ip dhcp snooping trust
exit
write memory
```

Server

ISP1

ISP-R1

```
en
conf t
host ISP-R1

int e1/1
 description Link_to_ISP-R4_Internal
 ip add 172.16.14.2 255.255.255.252
 no sh

int e1/2
 description Link_to_ISP-R3_Internal
 ip add 172.16.13.1 255.255.255.252
 no sh

int e0/2
 description Link_to_MB-R1
 ip add 203.0.113.1 255.255.255.252
 no sh

int e0/3
 description Link_to_RS-R2
 ip add 203.0.113.25 255.255.255.252
 no sh

int e1/0
 description BGP_to_ISP2-R1
 ip add 200.200.22.1 255.255.255.252
 no sh

exit



!—--------------------------------------------------------
!Rutare:

router eigrp 1
 network 172.16.13.0 0.0.0.3
 network 172.16.14.0 0.0.0.3
 network 203.0.113.0 0.0.0.3
 network 203.0.113.24 0.0.0.3
 passive-interface e1/0
 no auto
Exit
```

```
router bgp 6000
 neighbor 200.200.22.2 remote-as 7000
 neighbor 200.200.22.2 description ISP2-R2
  address-family ipv4
  neighbor 200.200.22.2 activate
  redistribute eigrp 1
 exit-address-family
exit

router eigrp 1
 redistribute bgp 6000 metric 10000 1 255 1 1500
exit
do write


ip route 0.0.0.0 0.0.0.0 200.200.22.2

router eigrp 1
 redistribute static metric 10000 1 255 1 1500
exit
```

ISP-R2

```
en
conf t
host ISP-R2

int e1/0
 description Link_to_ISP-R4_Internal
 ip add 172.16.24.2 255.255.255.252
 no sh

int e1/1
 description Link_to_ISP-R3_Internal
 ip add 172.16.23.1 255.255.255.252
 no sh

int e0/1
 description Link_to_EB-R1
 ip add 203.0.113.9 255.255.255.252
 no sh

int e0/2
 description Link_to_ML-R1
 ip add 203.0.113.13 255.255.255.252
 no sh

int e0/3
 description Link_to_CL-R2
 ip add 203.0.113.17 255.255.255.252
 no sh

exit



!—-------------------------------------------------------
!Rutare:


router eigrp 1
 network 172.16.23.0 0.0.0.3
 network 172.16.24.0 0.0.0.3
 network 203.0.113.8 0.0.0.3
 network 203.0.113.12 0.0.0.3
 network 203.0.113.16 0.0.0.3
 no auto
exit
```

ISP-R3

```
en
conf t
host ISP-R3

int e1/2
 description Link_to_ISP-R1_Internal
 ip add 172.16.13.2 255.255.255.252
 no sh

int e1/1
 description Link_to_ISP-R2_Internal
 ip add 172.16.23.2 255.255.255.252
 no sh

int e0/2
 description Link_to_RS-R1
 ip add 203.0.113.21 255.255.255.252
 no sh

exit



!—---------------------------------------------------------
!Rutare:

router eigrp 1
 network 172.16.13.0 0.0.0.3
 network 172.16.23.0 0.0.0.3
 network 203.0.113.20 0.0.0.3
 no auto
exit
```

ISP-R4

```
en
conf t
host ISP-R4

int e1/1
 description Link_to_ISP-R1_Internal
 ip add 172.16.14.1 255.255.255.252
 no sh

int e1/0
 description Link_to_ISP-R2_Internal
 ip add 172.16.24.1 255.255.255.252
 no sh

int e0/2
 description Link_to_MB-R2
 ip add 203.0.113.5 255.255.255.252
 no sh

exit



!—-------------------------------------------------------
!Rutare:


router eigrp 1
 network 172.16.14.0 0.0.0.3
 network 172.16.24.0 0.0.0.3
 network 203.0.113.4 0.0.0.3
 no auto
exit
```

ISP2

ISP2-R1

```
en
conf t
host ISP2-R1

int e0/0
 description Link_to_ISP2-R2_Internal
 ip add 172.16.51.1 255.255.255.252
 no sh

int e0/2
 description Link_to_ISP2-R3_Internal
 ip add 172.16.53.1 255.255.255.252
 no sh

exit


!—-------------------------------------------------------
!Rutare:


router eigrp 1
 network 172.16.51.0 0.0.0.3
 network 172.16.53.0 0.0.0.3
 no auto
Exit
```

ISP2-R2

```
en
conf t
host ISP2-R2

int e1/0
 description BGP_to_ISP-R1
 ip add 200.200.22.2 255.255.255.252
 no sh

int e0/0
 description Link_to_ISP2-R1_Internal
 ip add 172.16.51.2 255.255.255.252
 no sh

int e0/1
 description Link_to_ISP2-R3_Internal
 ip add 172.16.52.1 255.255.255.252
 no sh

exit


!—-------------------------------------------------------
!Rutare:

router eigrp 1
 network 172.16.51.0 0.0.0.3
 network 172.16.52.0 0.0.0.3
 passive-interface e1/0
 no auto
Exit

router bgp 7000
 neighbor 200.200.22.1 remote-as 6000
 neighbor 200.200.22.1 description ISP1-2
 address-family ipv4
  neighbor 200.200.22.1 activate
  network 0.0.0.0
  redistribute connected
  redistribute eigrp 1
 exit-address-family
exit

router eigrp 1
 redistribute bgp 7000 metric 10000 1 255 1 1500
exit
do write
```

ISP2-R3

```
en
conf t
host ISP2-R3

int e0/1
 description Link_to_ISP2-R2_Internal
 ip add 172.16.52.2 255.255.255.252
 no sh

int e0/2
 description Link_to_ISP2-R1_Internal
 ip add 172.16.53.2 255.255.255.252
 no sh

int e0/0
 description Link_to_Cloud_DHCP
ip address 172.17.0.50 255.255.0.0
 no sh

exit


!—-------------------------------------------------------
!Rutare:

ip route 10.50.0.0 255.255.0.0 172.17.0.60

ip route 0.0.0.0 0.0.0.0 172.17.0.1

router eigrp 1
 network 172.16.52.0 0.0.0.3
 network 172.16.53.0 0.0.0.3
redistribute static metric 10000 1 255 1 1500
 no auto
Exit


!____________________________________________
!Nat:


interface Ethernet0/1
 ip nat inside
interface Ethernet0/2
 ip nat inside
interface Ethernet0/0
 ip nat outside
exit
```

```
access-list 1 permit 10.0.0.0 0.255.255.255
access-list 1 permit 192.168.0.0 0.0.255.255
access-list 1 permit 172.16.0.0 0.15.255.255
access-list 1 permit 203.0.113.0 0.0.0.255
ip nat inside source list 1 interface Ethernet0/0 overload
```

```
! 1. Ștergem vechea regulă de NAT și vechiul ACL standard
no ip nat inside source list 1 interface Ethernet0/0 overload
no access-list 1

! 2. Creăm noul ACL extins (DENY = NU face NAT când pleci spre Datacenter)
access-list 100 deny ip 10.0.0.0 0.255.255.255 10.50.0.0 0.0.255.255
access-list 100 deny ip 192.168.0.0 0.0.255.255 10.50.0.0 0.0.255.255
access-list 100 deny ip 172.16.0.0 0.15.255.255 10.50.0.0 0.0.255.255
access-list 100 deny ip 203.0.113.0 0.0.0.255 10.50.0.0 0.0.255.255

! 3. Permitem NAT normal pentru ieșirea spitalului pe Internet
access-list 100 permit ip 10.0.0.0 0.255.255.255 any
access-list 100 permit ip 192.168.0.0 0.0.255.255 any
access-list 100 permit ip 172.16.0.0 0.15.255.255 any
access-list 100 permit ip 203.0.113.0 0.0.0.255 any

! 4. Reasociem NAT-ul cu noul ACL extins
ip nat inside source list 100 interface Ethernet0/0 overload

end
clear ip nat translation *
wr
```

Topologie_DataCenter

DT-R1

```
en
conf t
hostname DT-R1

! =============================================================
! 1. INTERFEȚE FIZICE
! =============================================================


interface Ethernet0/0
 description Link_to_Spital_via_GNS3_Cloud_Bridge
 ip address 172.17.0.60 255.255.0.0
 no shutdown
exit

interface Ethernet0/1
 description Transit_Link_to_DC_Core_Switch
 ip address 10.50.1.1 255.255.255.252
 no shutdown
exit

! =============================================================
! 2. RUTE STATICE (RUTARE LAYER 3)
! =============================================================

! Direcționăm tot traficul pentru rețelele spitalului prin nor,
! trimițând pachetele către IP-ul lui ISP2-R3 (172.17.0.50)
ip route 10.1.0.0 255.255.0.0 172.17.0.50
ip route 10.2.0.0 255.255.0.0 172.17.0.50
ip route 10.3.0.0 255.255.0.0 172.17.0.50
ip route 10.4.0.0 255.255.0.0 172.17.0.50
ip route 192.168.0.0 255.255.0.0 172.17.0.50


ip route 10.50.0.0 255.255.0.0 10.50.1.2
 ip route 0.0.0.0 0.0.0.0 172.17.0.50

ip route 203.0.113.0 255.255.255.0 172.17.0.50

ip route 10.0.0.0 255.255.0.0 172.17.0.50
```

```
!_______________________________________
!Nat:

interface Ethernet0/1
 ip nat inside
exit

interface Ethernet0/0
 ip nat outside
exit



access-list 100 deny ip 10.50.0.0 0.0.255.255 10.0.0.0 0.255.255.255
access-list 100 deny ip 10.50.0.0 0.0.255.255 192.168.0.0 0.0.255.255
access-list 100 deny ip 10.50.0.0 0.0.255.255 172.16.0.0 0.15.255.255
access-list 100 deny ip 10.50.0.0 0.0.255.255 172.17.0.0 0.0.255.255
access-list 100 deny ip 10.50.0.0 0.0.255.255 203.0.113.0 0.0.0.255
access-list 100 permit ip 10.50.0.0 0.0.255.255 any

access-list 100 deny ip 10.1.0.0 0.0.255.255 10.50.0.0 0.0.255.255
access-list 100 permit ip 10.1.0.0 0.0.255.255 any

ip nat inside source list 100 interface Ethernet0/1 overload

end
clear ip nat translation *
wr

configure terminal
no ip nat inside source list 100 interface Ethernet0/1 overload
ip nat inside source list 100 interface Ethernet0/0 overload
write memory
```

DT-SW-CORE

```
en
conf t
hostname DT-SW-CORE

! --- 1. Activare Rutare L3 și Rapid Spanning Tree ---
ip routing
spanning-tree mode rapid-pvst
spanning-tree vlan 5,10,20,30,40 priority 4096

! --- 2. Definire Bază de Date VLAN-uri ---
vlan 5
 name Management
vlan 10
 name Web_Servers
vlan 20
 name Database_Servers
vlan 30
 name App_Servers
vlan 40
 name Backup_Storage
vlan 99
 name Native
exit

! --- 3. Legătură Downstream către DT-SW1 (Port-channel 1) ---
interface range Ethernet1/0 - 1
 description Link_to_DT-SW1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 channel-group 1 mode active
 no shutdown
exit

interface Port-channel1
 description Trunk_to_DT-SW1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 no shutdown
exit

! --- 4. Legătură Downstream către DT-SW2 (Port-channel 2) ---
interface range Ethernet1/2 - 3
 description Link_to_DT-SW2
 switchport trunk encapsulation dot1q
```

```
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 channel-group 2 mode active
 no shutdown
exit

interface Port-channel2
 description Trunk_to_DT-SW2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 no shutdown
exit

! --- 5. Configurare SVI-uri (Default Gateways pentru VLAN-uri) ---
interface Vlan5
 description Gateway_Management
 ip address 10.50.5.1 255.255.255.0
 no shutdown

interface Vlan10
 description Gateway_Web_Servers
 ip address 10.50.10.1 255.255.255.0
 no shutdown

interface Vlan20
 description Gateway_Database_Servers
 ip address 10.50.20.1 255.255.255.0
 no shutdown

interface Vlan30
 description Gateway_App_Servers
 ip address 10.50.30.1 255.255.255.0
 no shutdown

interface Vlan40
 description Gateway_Backup_Storage
 ip address 10.50.40.1 255.255.255.0
 no shutdown

! --- 6. Pregătire Port către Ruterul de Margine ---
interface Ethernet0/1
 description Link_to_DT-R1
 no switchport
 ip address 10.50.1.2 255.255.255.252
 no shutdown
```

```
!—-------------------------------------------------------
! Rutare:
ip route 0.0.0.0 0.0.0.0 10.50.1.1
do write
```

DT-SW1

```
en
conf t
hostname DT-SW1

no ip routing
spanning-tree mode rapid-pvst

! --- 1. Definire VLAN-uri Datacenter ---
vlan 5
 name Management
vlan 10
 name Web_Servers
vlan 20
 name Database_Servers
vlan 30
 name App_Servers
vlan 40
 name Backup_Storage
vlan 99
 name Native
exit

! --- 2. Porturi de Access (Conform Topologiei din GNS3) ---
interface Ethernet0/0
 description Access_PC1_VLAN10
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

interface Ethernet0/1
 description Access_PC2_VLAN20
 switchport mode access
 switchport access vlan 20
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

! --- 3. Pregătire Interfețe Trunk (Fizice) ---
interface range Ethernet1/0 - 1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 channel-group 1 mode active
 no shutdown
exit
```

```
interface Port-channel1
 description Trunk_to_DT-SW-CORE
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 no shutdown
exit

! --- 5. IP de Management (VLAN 5) ---
interface Vlan5
 description Management_IP
 ip address 10.50.5.11 255.255.255.0
 no shutdown
exit

ip default-gateway 10.50.5.1
do write
```

!___________________________________
!Port security:

```
configure terminal
ip dhcp snooping
no ip dhcp snooping information option
ip dhcp snooping vlan 10,20

interface range Ethernet0/0 - 1
 switchport port-security
 switchport port-security maximum 2
 switchport port-security violation shutdown
 switchport port-security mac-address sticky
 ip dhcp snooping limit rate 15

interface Port-channel1
 ip dhcp snooping trust
exit
write memory
```

DT-SW2

```
en
conf t
hostname DT-SW2

no ip routing
spanning-tree mode rapid-pvst

! --- 1. Definire VLAN-uri Datacenter ---
vlan 5
 name Management
vlan 10
 name Web_Servers
vlan 20
 name Database_Servers
vlan 30
 name App_Servers
vlan 40
 name Backup_Storage
vlan 99
 name Native
exit

! --- 2. Porturi de Access (Conform Topologiei din GNS3) ---
interface Ethernet0/0
 description Access_PC3_VLAN30
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

interface Ethernet0/1
 description Access_PC4_VLAN40
 switchport mode access
 switchport access vlan 40
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

! --- 3. Pregătire Interfețe Trunk (Fizice) ---
interface range Ethernet1/2 - 3
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 channel-group 1 mode active
 no shutdown
exit
```

```
interface Port-channel1
 description Trunk_to_DT-SW-CORE
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 5,10,20,30,40,99
 no shutdown
exit

! --- 5. IP de Management (VLAN 5) ---
interface Vlan5
 description Management_IP
 ip address 10.50.5.12 255.255.255.0
 no shutdown
exit

ip default-gateway 10.50.5.1
do write
```

!____________________________________

!Port security:

```
configure terminal
ip dhcp snooping
no ip dhcp snooping information option
ip dhcp snooping vlan 30,40

interface range Ethernet0/0 - 1
 switchport port-security
 switchport port-security maximum 2
 switchport port-security violation shutdown
 switchport port-security mac-address sticky
 ip dhcp snooping limit rate 15

interface Port-channel1
 ip dhcp snooping trust
exit
write memory
```

NTP/Syslog

```
ip a    # vezi numele interfeței (ens3)
sudo nano /etc/netplan/01-network-manager-all.yaml
```

```yaml
network:
  version: 2
  renderer: NetworkManager
  ethernets:
    ens3:
      dhcp4: no
      addresses:
        - 10.50.30.10/24
      routes:
        - to: default
          via: 10.50.30.1
      nameservers:
        addresses: [8.8.8.8]
```

```
sudo netplan apply
```

Install NTP:

```
sudo apt update
sudo apt install chrony -y

sudo nano /etc/chrony/chrony.conf
```

```
# Permite clienților din rețeaua spitalului și datacenter
allow 10.0.0.0/8
allow 192.168.0.0/16
allow 203.0.113.0/24

# Sincronizare cu pool real (serverul are internet)
pool pool.ntp.org iburst
```

```
sudo systemctl restart chrony
sudo systemctl enable chrony

chronyc tracking
sudo ss -ulnp | grep 123

sudo ufw allow 123/udp
```

Install Syslog:

sudo nano /etc/rsyslog.conf

Decomentat:

```
module(load="imudp")
input(type="imudp" port="514")
```

---

sudo nano /etc/rsyslog.d/10-cisco.conf

```
sudo tee /etc/rsyslog.d/10-cisco.conf > /dev/null << 'EOF'
template(name="RemoteLogs" type="string"
string="/var/log/cisco/%FROMHOST-IP%/messages.log")

if ($fromhost-ip startswith '10.' or $fromhost-ip startswith '203.0.113.' or $fromhost-ip
startswith '192.168.') then {
   action(type="omfile" dynaFile="RemoteLogs" dirCreateMode="0755"
fileCreateMode="0644")
   stop
}
EOF
```

```
sudo mkdir -p /var/log/cisco
sudo systemctl restart rsyslog

sudo ufw allow 514/udp
sudo ss -ulnp | grep 514

sudo chown -R syslog:syslog /var/log/cisco
sudo chmod 755 /var/log/cisco
```